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Wake-supervised coefficient states for wall-pressure estimation of extreme vortex-gust airfoil encounters

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Extreme vortex-gust encounters of a post-stall airfoil are wake-reorganisation events: the transient load is the surface signature of leading-edge-vortex roll-up and shedding, and the release parameters alone do not fix the wake once the gust has passed. A reduced-order model of such an encounter is useful in deployment only if its state can be recovered from the sensors actually available, the wall pressure, and advanced without leaving the region on which that recovery was calibrated. Using direct numerical simulations of a NACA 0012 airfoil at $\alpha = 14^\circ$ and $Re = 5000$, perturbed by Taylor vortices spanning gust strength, core diameter and wall-normal offset, we ask which reduced coefficient state a sequential wall-pressure estimator can track through the encounter. At fixed dimension, under one shared probe and predictor, we compare a wake-supervised predictive joint-embedding state, a reconstructive autoencoder in the Fukami-Taira lineage and a proper orthogonal decomposition basis, with matched controls separating objective from supervision. We find a division of labour: the supervision supplies the wake's instantaneous readability and an anisotropic latent geometry the forward rollout respects, while the multi-step predictive objective supplies long-horizon rollout stability. A leakage-free ensemble Kalman filter sensing eight wall-pressure taps tracks the lift (analysis R^2 of 0.74 to 0.84 for gust ratios of unity and above) where every static inverse fails by a gust ratio of three; run identically on each family, the filter tracks the load through any nonlinear latent, and the predictive state is distinguished at the strongest gusts, in calibration, and in carrying a wake-readable state. One tap read over a thirty-frame window recovers the state as well as eight taps at an instant, a trade read through the delay-embedding theorem, and the envelope closes where the strengthening gust drives the interaction three-dimensional. The filter is under-dispersed at the strongest gusts, where a calibrated noise model, rather than more sensors, is the outstanding requirement.

Key words: vortex shedding, low-dimensional models, machine learning

1. Introduction

Small uncrewed aerial vehicles are increasingly expected to operate in environments where the incoming flow is neither steady nor weakly disturbed. Urban canyons, mountainous terrain, coastal regions and ship or building wakes can expose these vehicles to gusts whose characteristic length scale is comparable to the wing chord and whose velocity is of the same order as the flight speed (Jones *et al.* 2022; Mohamed *et al.* 2023; Fukami & Taira 2023). In this regime, gusts cannot be treated as small perturbations of an otherwise steady aerodynamic state. They interact with separation, vortex shedding and wake recovery, producing large transient loads and rapid changes in the surrounding flow organisation. For small vehicles, this coupling directly affects navigability, control authority and flight safety.

This regime has motivated a growing body of work on extreme aerodynamics (Taira 2026). A canonical example is the interaction between a discrete vortex gust and a wing or airfoil, where the disturbance has sufficient strength to reorganise the near wake rather than merely modulate the instantaneous force (Fukami *et al.* 2025). Recent studies have shown that apparently complex gust encounters can exhibit unexpectedly low-dimensional structure when viewed in suitable coordinates. Lift-augmented autoencoders have revealed compact manifolds for extreme vortex-gust airfoil interactions (Fukami & Taira 2023), and the same reduced description has been used to support phase-amplitude analysis and data-driven transient lift attenuation (Fukami *et al.* 2024). Related work has also shown that large-scale vortical structures can display similarities across laminar and turbulent extreme-gust encounters (Odaka *et al.* 2026). These results suggest that the dynamical organisation of extreme aerodynamic flows may be simpler than the full flow-field dimension implies.

For practical deployment, however, compactness alone is not sufficient. A reduced description intended for gust mitigation must be usable as a plant state: it must be estimated from available sensors, advanced in time by a predictive model and queried by a controller or decision algorithm (Loiseau *et al.* 2018). This requirement places the problem closer to state estimation and model-based control than to flow-field compression alone. Surface pressure is especially important in this context, because it is one of the few measurements that can be obtained at high bandwidth on an aerodynamic body. Recent gust-modelling studies have therefore used sparse surface-pressure histories to infer unsteady aerodynamic loads and, more broadly, to bridge spatial pressure correlations and temporal flow memory (Zhong *et al.* 2023; Haughn *et al.* 2024; Chen *et al.* 2026), and Bayesian sequential estimation has begun to close the loop between sparse pressure and low-order aerodynamic states, with uncertainty carried alongside the estimate (Mousavi & Eldredge 2025; Eldredge & Mousavi 2025). Such approaches indicate a path towards deployable gust-response models, but they also sharpen the central question: what reduced state should be estimated from the wall in the first place?

Most current nonlinear reduced-order models for extreme aerodynamic flows are built around reconstruction-trained autoencoders. In this approach, a convolutional encoder maps the flow field to a low-dimensional latent vector, and a decoder reconstructs the original field. Aerodynamic observables, such as lift, can be added as auxiliary losses to orient the latent space towards physically relevant quantities (Fukami & Taira 2023, 2025). This strategy is powerful because it preserves a direct connection between the latent variables and the resolved flow. It also provides a natural route to visualisation and

interpretation, since the decoder maps latent coordinates back to physical space. Similar ideas have been combined with variational autoencoders and sequence models to build predictive reduced-order models of chaotic flows (Solera-Rico *et al.* 2024; Fukagata & Fukami 2025).

Yet a reconstruction-trained latent space is not necessarily the best space for prediction. The decoder encourages the encoder to preserve information needed to reproduce the instantaneous field, but the time-advancing model requires coordinates that remain stable and informative under autoregressive rollout. These two requirements need not coincide. Recent comparisons in controlled wake flows indicate that convolutional-autoencoder-based predictors can be more prone to long-horizon divergence, whereas POD-based predictors, despite their linear and less compact representation, may retain higher median forecast accuracy and a narrower spread (Solera-Rico *et al.* 2025). This does not make POD preferable as a general representation of nonlinear separated flows. Rather, it exposes a tension: nonlinear encoders can compress more aggressively, but reconstruction compactness does not guarantee forecast robustness. The question is therefore whether one can construct nonlinear reduced states that retain the compactness of learned manifolds while improving their predictive usability.

This question motivates the use of a joint-embedding predictive architecture (JEPA) (LeCun 2022). In contrast with an autoencoder, a JEPA does not train the representation by reconstructing the input field. Instead, it learns an encoder and a predictor such that the present latent state predicts a future latent state. In computer vision, image- and video-based JEPAs have been introduced as non-generative self-supervised models that predict in representation space rather than pixel space (Assran *et al.* 2023, 2025). For the present fluid-mechanics problem, the relevance of JEPA is more specific: it provides a way to train the reduced coordinates in the same space in which the forward model will later operate. The architecture is therefore used here as a diagnostic tool to test whether predictive training can produce a nonlinear state that is more stable under rollout than a reconstruction-trained latent.

This choice creates a second question. Because the JEPA state is trained without a decoder, it is not obvious what physical information is retained by its coordinates. A low forecast error in latent space would be insufficient if the state did not carry the wake variables that matter for gust response. We therefore treat decoding not as part of the training objective, but as an a-posteriori diagnostic. After training the reduced state, we ask what can be recovered from it: aerodynamic observables, wake enstrophy, flow-field structures and, ultimately, the latent coordinates themselves from sparse wall pressure. This separation between predictive training and diagnostic decoding allows us to distinguish three properties that are often conflated: whether a state is compact, whether it is physically readable and whether it remains usable when advanced by its own dynamics.

The present work examines these questions using a canonical post-stall vortex-gust interaction. The gust parameters are not supplied to the predictive model; the encoder receives only the instantaneous flow state, and the predictor advances the latent state from its own history. We compare a predictive joint-embedding model with reconstruction-trained nonlinear encoders and a linear POD representation at matched latent dimension. The comparison is not designed to identify the method that reconstructs the vorticity field most accurately. Instead, it asks which representation supplies a reduced aerodynamic state that can be forecast, decoded and estimated in a form relevant to future gust-mitigation systems.

The paper makes three contributions. First, we pose the extreme vortex-gust encounter as a predictive-state estimation problem and fix its endpoints in advance: the reduced state must carry the post-impact wake response, summarised by the wake enstrophy and

complementary flow diagnostics that scalar force histories alone can miss, and it must be recoverable from wall pressure and advanced by a forward model without leaving the region on which that recovery was fitted. Second, we separate the ingredients of such a state with matched controls, and reach a division of labour. Observable supervision on the wake supplies both its instantaneous readability and an anisotropic latent geometry into which the forward rollout settles, where an anti-collapse regulariser applied to a bare reconstruction leaves the latent near-isotropic and its rollout departs along low-variance directions; the multi-step predictive objective supplies the long-horizon rollout stability that neither a single-step objective nor an objective-free supervised encoder provides. Latent drift is the mechanism through which an apparently accurate reconstructive state loses this physics over the rollout. Third, we demonstrate the state in a leakage-free sequential filter that senses only wall pressure and never observes the vorticity field, and we map the gust-intensity envelope over which it holds. The filter tracks the transient load through the encounter and extends the usable gust range beyond what any static inverse achieves, and a delay-coordinate reading explains why: the encounter is low-dimensional, so a short window of pressure reconstructs the state a single frame cannot, and the envelope closes where the strengthening gust drives the interaction three-dimensional and its effective dimension outruns a fixed sensing budget. We are explicit about the limits: run identically on every family, the filter tracks the load through any nonlinear latent, and the predictive state's distinctions are the strongest gusts, the calibration margin, and the wake content of the tracked state; its recoverability margin over a raw field latent is modest; and the filter is under-dispersed at the strongest gusts. This study investigates a fluid-mechanics question through a learning architecture: what kind of reduced state can carry the wake dynamics of an extreme gust encounter while remaining estimable from the wall and predictive enough to serve as a future plant model?

2. Flow configuration and data

2.1. Vortex gust-airfoil interaction

We study a NACA 0012 airfoil at angle of attack $\alpha = 14^\circ$ and chord Reynolds number $Re = 5000$, perturbed by a discrete vortex gust modelled as a spanwise-oriented Taylor vortex released upstream of the leading edge. The data analysed here are our own direct numerical simulations of this configuration, computed with the GPU-enabled spectral-element solver SOD2D (Gasparino *et al.* 2024) with no subgrid-scale model, so that all scales of the separated wake are resolved at this Reynolds number. The solver is run in the low-Mach regime in which the flow is effectively incompressible. The configuration and its physical characterisation follow Fukami, Smith & Taira (2025), which is the reference for the physics, not the source of the data. The vortex has the azimuthal velocity profile

$$u_\theta(r) = u_{\theta,\max} \frac{r}{R} \exp\left[\frac{1}{2}\left(1 - \frac{r^2}{R^2}\right)\right], \quad (2.1)$$

which peaks at the core radius $r = R$ (Taylor 1918), and is parametrised by three scalars: the signed gust ratio $G \equiv u_{\theta,\max}/u_\infty$, the core diameter $D \equiv 2R/c$, and the wall-normal offset Y of the vortex centre, the displacement normal to the chord (the laboratory coordinates are rotated by the angle of attack to define Y). The released-data case identifiers encode the solver's opposite sign convention, $s = -G$, so a case labelled G+4.00 in the archive corresponds to $G = -4$ here. The profile and the (G, D, Y) sampling follow the configuration of Fukami *et al.* (2025). The training envelope spans $|G| \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ with $G = 0$ as the undisturbed reference, $D \in \{0.5, 1.0, 1.5\}$

chords, and $Y \in [-0.4, +0.4]$; the value $|G| = 4$ is held out as an extrapolation boundary, for reasons that are physical and that we return to below. Convective time t is measured in chords and set to zero when the vortex centre reaches the leading edge.

Without a gust, the $\alpha = 14^\circ$ wake at this Reynolds number is massively separated and sheds periodically; in dynamical terms the undisturbed flow is a limit cycle. A gust encounter is then a transient excursion from this limit cycle and back, and it is helpful to read it in stages (figure 1) (Smith *et al.* 2024; Fukami *et al.* 2025; Odaka *et al.* 2026). The vortex first induces a strong wall-normal vorticity flux at the leading edge, the airfoil experiences a sharp lift excursion as a leading-edge vortex (LEV) forms, the LEV rolls up and sheds together with a trailing-edge structure as the load reaches its extremum, and the wake then relaxes back toward the baseline shedding. The sign of G sets whether the first lift excursion is positive or negative, Y sets the side and strength of the impingement, D sets how much of the encounter is dominated by the vortex core, and the impact timing sets the phase of the baseline cycle at which the disturbance arrives. The five observables defined in §2.2 are quantitative summaries of exactly this sequence.

A consequence of the source characterisation (Fukami *et al.* 2025) is central to how we read the extrapolation case. For $|G| \leq 3$ the encounter is primarily two-dimensional, so a mid-plane slice of the spanwise vorticity carries the relevant large-scale dynamics; for $|G| \geq 4$ the interaction becomes genuinely three-dimensional, with fine-scale instabilities developing during impact. We confirm this from the full three-dimensional vorticity fields: the fraction of the spanwise-vorticity (ω_z) enstrophy not captured by its spanwise mean, the content a mid-plane slice discards, is roughly constant across the training range ($|G| \leq 3$, post-impact wake median ≈ 0.20) and rises about threefold at $|G| = 4$ (≈ 0.56 , over the four available cases). Because the encoder in this work consumes a single mid-plane vorticity field (§2.2), the held-out $|G| = 4$ case is not merely one parameter step beyond the training range; it is a regime in which the quantity the encoder observes no longer determines the true dynamics. We therefore treat $|G| = 4$ as a physical observability boundary rather than as a claim of parametric generalisation.

2.2. Data, fields, and observables

Each simulation case is one long run partitioned into encounters of 120 cached frames at $\Delta t = 0.05 t/c$, one encounter per gust release. The simulations use a spectral-element discretisation of polynomial order $p = 4$, and the fields are cached on a structured grid of $192 \times 96 \times 32$ points spanning $x/c \in [-1.5, 4.5]$ and $y/c \in [-1.5, 1.5]$ with a spanwise extent of approximately one chord; this analysis subdomain matches the data-driven subdomain of the source characterisation (Fukami *et al.* 2025). The lift and drag coefficients are $C_L = F_L / (\frac{1}{2} \rho u_\infty^2 c)$ and $C_D = F_D / (\frac{1}{2} \rho u_\infty^2 c)$, obtained by integrating the pressure and viscous stresses over the airfoil surface. The remaining DNS solver-resolution details required for full reproducibility, the free-stream Mach number, the domain and spanwise extent, the element and solution-point counts, the near-wall spacing, the time step and CFL, the gust-release station, and the grid and time-step sensitivity, are listed as an author-fill checklist in table 1. The dataset comprises 102 cases partitioned into training, validation (the held-out encounters of the training cases, denoted test_a), an in-distribution test set (test_b), and an out-of-distribution test set (test_c). The in-distribution test set is stratified to span the gust strengths, the three core diameters, and the two source groups (figure 2); test_c is the $|G| = 4$ extrapolation set, sampled symmetrically in both signs of the gust (§2.2). These 102 cases are separated in preprocessing into 450 encounter windows of 120 frames each, one per gust release: 268 training, 100 validation (test_a, the held-out last encounters of the training cases), 42 test_b, and 40 test_c. The undisturbed reference case is retained in training. Behind these encounter counts are far fewer distinct cases:

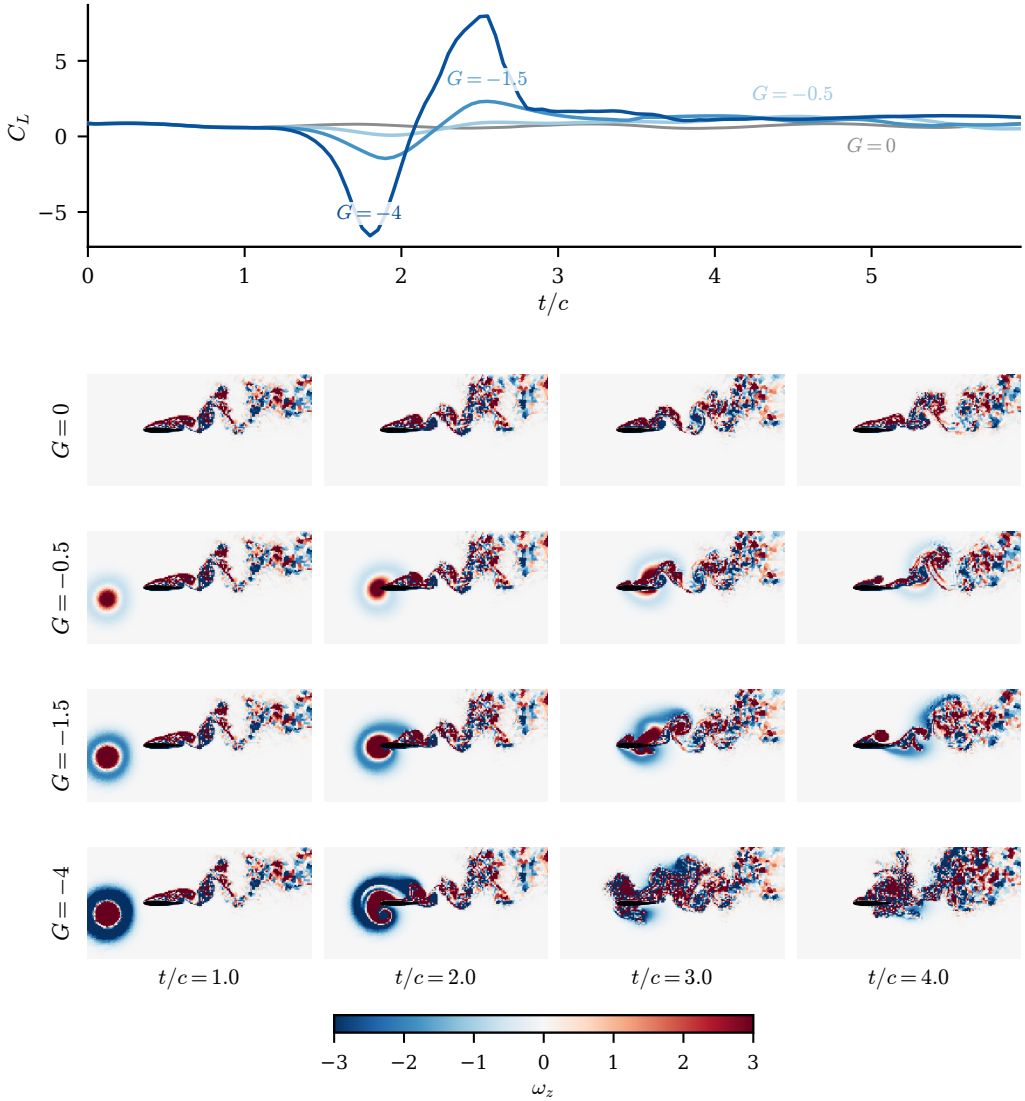


Figure 1. The vortex gust as a wake-reorganisation event, shown as a gust-strength sweep at fixed core diameter $D = 0.5$ and wall-normal offset $Y = -0.1$ (the first encounter of each case). (a) The lift coefficient C_L against convective time t/c for the four cases, with $t/c = 0$ the gust-release instant and impact near $t/c = 2$; the undisturbed baseline ($G = 0$) is grey, and the load excursion deepens and widens monotonically with the gust strength $|G|$. (b) The mid-plane spanwise vorticity ω_z for each case (rows, $G = 0, -0.5, -1.5, -4$) at four convective times (columns, $t/c = 1, 2, 3, 4$), in raw physical units with the colourbar clipped at ± 3 (the intense cores saturate; the wake structure away from them lies within the range), and the airfoil drawn as a filled polygon. The strong gust injects a vortex that reorganises the wake through impact, while the weak cases perturb the baseline shedding only slightly; $G = -4$ is the $|G| = 4$ extrapolation case, shown here only as a physical illustration. The load spike and the wake reorganisation are the surface and field signatures of the same event, and the release parameters alone do not fix where the wake sits afterwards, which is why the wake, not only C_L , is the endpoint a reduced state must carry.

Quantity	Value	Why a referee needs it
Free-stream Mach number (or incompressible-limit confirmation)	[PENDING:]	Confirms the low-Mach formulation is effectively incompressible at $Re = 5000$.
Computational domain and spanwise extent L_z/c	[PENDING:]	Bounds domain blockage and confirms the span resolves the three-dimensional wake.
Element and solution-point counts	[PENDING:]	Fixes the spatial resolution of the spectral-element discretisation.
Minimum wall-normal spacing $\Delta n_{\min}/c$ and wall units Δn^+	[PENDING:]	Shows the near-wall layer is resolved without a wall model.
Time step $\Delta t u_\infty/c$ and maximum CFL	[PENDING:]	Establishes the temporal resolution and stability of the integration.
Gust-release station x_0/c	[PENDING:]	Sets the upstream distance over which the Taylor vortex develops before impact.
Grid and time-step sensitivity check (baseline and a representative gust case)	[PENDING:]	Demonstrates that the reported statistics are converged.

Table 1. DNS solver-resolution details required for full reproducibility (§2.2). The data are direct numerical simulations with no subgrid-scale model, so all scales are resolved and no large-eddy closure is involved. Values marked pending are owned by the simulation collaborators and are filled from `paper/dns_metadata.yaml` before submission.

the 42 test_b encounters come from only 10 held-out cases (6 interior and 4 boundary tier) and the 40 test_c encounters from 8 cases, with most cases contributing four to six encounters that share a baseline shedding history. Encounters of the same case are therefore not independent, so encounter-level intervals overstate the effective sample size; the headline comparisons are reported additionally with a case-clustered bootstrap and a case-level paired test (Appendix A). The partition is fixed once and used identically by every model and baseline.

The $|G| = 4$ extrapolation boundary is sampled symmetrically by combining the periodic archive cases with four additionally released cases at the opposite sign of the gust. We verified the additional cases against the archive by a physical consistency check on the load rather than by trusting the file metadata: at matched core diameter and offset, the two signs of the $|G| = 4$ gust produce near mirror-image lift transients, one a positive-first excursion peaking near $+8.3$ and the other troughing near -7.6 , as the sign of the gust ratio sets the sign of the first excursion (§2.1). This confirms that the two halves of the symmetric boundary set are physically consistent and that the sign-asymmetry analysis of the operating envelope (§4.5) is sound. Sampling the boundary at both signs is a genuine improvement over a one-sided $|G| = 4$ set: it lets the sign asymmetry be measured rather than assumed.

For each frame the cache stores the mid-plane spanwise vorticity $\omega_z(192, 96)$, the span-averaged wall pressure $p_{\text{wall}}(192)$, and the lift and drag coefficients C_L, C_D . The vorticity is normalised by a fixed pipeline: a spatial mask removes the solid and one adjacent cell layer to suppress the leading-edge finite-difference artefact, a per-encounter high-percentile clip caps spikes, and the field is scaled by three times the training standard deviation ($\sigma_{\text{train}} = 3.5396$) with no mean shift, preserving the vorticity antisymmetry. All representation and probe losses are computed in this normalised space; unnormalised values are used only for physical-units metrics and figures. The field-reconstruction similarity reported below uses the structural similarity index of Wang *et al.* (2004) with the standard constants $K_1 = 0.01$, $K_2 = 0.03$ and a dataset-dependent dynamic range $L = 8.487$, set to twice the global

99.9th percentile of the normalised target magnitude over the validation split so that the index is comparable across reruns. Appendix A repeats the primary wake probe under no amplitude clip and under a single training-pooled leak-free threshold (established on the previous-generation encoders; the pipeline is unchanged in this revision), and the family ordering is unchanged, so the per-encounter clip does not drive the result.

We evaluate forward closure against five physical observables that together span body-force and spatially distributed wake diagnostics. The lift and drag coefficients C_L and C_D are the body forces. The wake enstrophy and the positive and negative circulation are spatially distributed wake observables: they integrate (signed) vorticity over the wake region and are the most demanding to forecast, because they are sensitive to the redistribution of vorticity that the LEV roll-up and shedding produce. The choice is deliberate: the wake observables can change sharply while the scalar forces change little, so they are the part of the encounter most likely to separate a representation that has captured the gust-vortex physics from one that has only captured its force signature. On these grounds we fix two co-primary endpoints in advance: the wake enstrophy, which the reduced state must carry beyond the release parameters, and the lift coefficient C_L , which the sequential wall-pressure estimator of §3.3 tracks and in which the operating envelope is stated. The remaining three observables are reported as secondary and descriptive. This pre-registration is what the family-wide multiplicity correction in Appendix A is applied against.

The five observables are defined on each frame of the masked vorticity field ω_z described above. The lift and drag coefficients C_L, C_D are as defined earlier in this section. On the wake window $\Omega_w = \{(x, y) : x/c \in [0.5, 4], |y/c| \leq 1\}$ the wake enstrophy is

$$E_w(t) = \int_{\Omega_w} \omega_z^2 dA, \quad (2.2)$$

and the signed wake circulations integrate the vorticity over its strongly rotational parts,

$$\Gamma^+(t) = \int_{\Omega_w, \omega_z > \omega_c} \omega_z dA, \quad \Gamma^-(t) = \int_{\Omega_w, \omega_z < -\omega_c} \omega_z dA, \quad (2.3)$$

with threshold $\omega_c = 1$ in the cleaned vorticity units. The signed circulations at $\omega_c \in \{0.5, 1, 2\}$ are nearly collinear (pairwise Pearson well above 0.9 across held-out frames) and the predictive latent’s circulation closure advantage over the reconstructive baseline is essentially unchanged across them, so the closure does not depend on this threshold. Every case uses the same fixed window and mask, and values are reported in the cache vorticity units integrated over chord-normalised area. The forecast horizon $H = 16$ is the sixteenth cached frame after impact, $t = 16 \Delta t = 0.8 c/u_\infty$.

One definitional point bounds how these observables are read. The wake and flow observables (the enstrophy and the two circulations) are computed on the same mid-plane slice the encoder consumes, so they omit the out-of-plane content quantified in §2.1 (roughly a fifth of the ω_z enstrophy in distribution), whereas C_L and C_D are true three-dimensional surface-integrated forces. The closure therefore compares mid-plane wake diagnostics against three-dimensional forces, and the mid-plane proxy is weakest at the $|G| = 4$ extrapolation boundary, where the out-of-plane fraction rises substantially.

3. Predictive latent representation and evaluation protocol

3.1. Model and training objective

The model takes no gust parameters: neither the encoder nor the predictor receives (G, D, Y) . The predictive encoder E_θ maps the mid-plane vorticity field $\omega_z(192, 96)$ to a

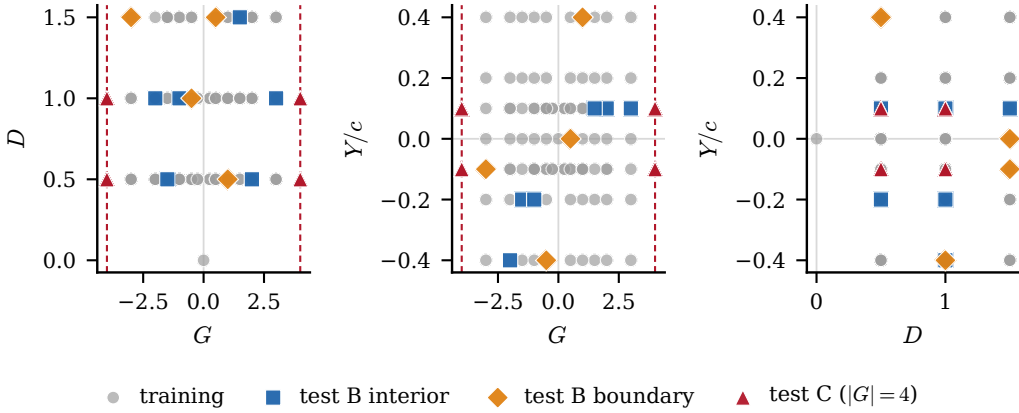


Figure 2. Parameter-space sampling of the 102 cases in the (G, D, Y) gust envelope, shown as three two-dimensional projections and coloured by split: training (268 encounters), the stratified in-distribution test set (test_b, 42 encounters from 10 cases, split into interior and boundary tiers), and the out-of-distribution extrapolation set (test_c, 40 encounters from 8 cases). The dashed lines mark the $|G| = 4$ extrapolation boundary, now sampled at both signs of the gust so that the boundary set is symmetric rather than one-sided. Training mass concentrates near $Y/c = 0$, which is why the wall-normal offset is the least well resolved parameter (§4.6).

latent $\mathbf{z} \in \mathbb{R}^d$ at the headline dimension $d = 32$. It is a hybrid network: three convolutional downsampling stages produce a $24 \times 12 \times 256$ feature map, a six-layer vision transformer processes the resulting tokens, and the class-token output is projected to $\mathbf{z}$ through a single linear layer with a batch-normalisation boundary, which the characteristic-function regulariser below requires. The encoder is therefore a pure state map, and this is the property that lets the latent itself serve as the observable a controller would have access to, since at deployment the gust parameters are not known a priori (§5).

The predictor P_ϕ advances the pooled state in time, from its own history alone, so the model is unconditioned end to end. It is a vector sequence model: a six-layer autoregressive transformer with rotary temporal positions and a causal mask that maps a short history of 32-vectors to the next 32-vector, with no gust input. From the two most recent encoded states it rolls eight steps open loop, feeding its own predictions back in, and the loss is the multi-step error between the rolled coefficients and the online encoder outputs at the matching frames, detached. The pipeline is therefore as plain as a linear or autoencoder reduced-order model: 32 numbers in, a dynamics model on the vector, 32 numbers out, with no spatial map anywhere in the loop. The state is the pooled vector at every stage: the anti-collapse regulariser, the observable heads, the probes, and the estimator all read the same 32-vector, and nothing in the model carries more. The estimator of §3.3 refits a predictor of this same class on the frozen pooled trajectories for its forecast step, so the training and estimation dynamics share a function class. Internal structure for both is given in Appendix A. Figure 3(a) shows the encoder and predictor, and panel (b) sets out the matched-predictor evaluation protocol used throughout; the contrast between the predictive training signal and the reconstructive one, in which the reconstructive autoencoder passes the field at time t through an encoder and a decoder and incurs a field-space error whereas the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss, is drawn in Appendix A (figure 11). Table 2 summarises the two networks.

The predictive encoder and predictor are trained jointly on a multi-step open-loop rollout term, the predictor rolled eight steps from its own outputs with no teacher forcing against

Table 2. Encoder and training-predictor configuration, counted from the released $d = 32$ checkpoint. The model is unconditional: no gust parameters $\mathbf{c} = (G, D, Y)$ enter the encoder or the predictor. Two small auxiliary heads read observables off the encoder output during training (a lift head and an 80-dimensional wake-spectrum head; §3.1). The estimation-stage forecast model of §3.3 is a predictor of the same class refitted on the frozen pooled trajectories.

	Encoder E_θ	Predictor P_ϕ
role	field $\rightarrow$ pooled state (unconditional)	pooled-state rollout (unconditional)
input	$\omega_z \in \mathbb{R}^{192 \times 96}$	pooled states $\mathbf{z} \in \mathbb{R}^{32}$
backbone	3 conv. stages + 6-layer ViT	6-layer causal transformer (RoPE)
width / heads	256 / 8	384 / 16
image tokens / context	$24 \times 12 = 288$	2 frames, rolled 8 steps open loop
conditioning	none	none
state read-out	class token $\rightarrow$ linear $\rightarrow$ BatchNorm	next pooled state
parameters	6.7M	10.7M

detached online targets, and an anti-collapse regulariser that prevents the encoder from collapsing to a constant. We use the characteristic-function regulariser of Balestrierio & LeCun (2025); Maes *et al.* (2026), which matches the latent distribution to an isotropic Gaussian through a small number of random projections, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) should the participation ratio of the latent indicate collapse. Its configuration is given in Appendix A.

Two auxiliary heads read aerodynamic observables off the encoder output during training: a lift head that maps $\mathbf{z}_t$ to $C_L(t)$, and a wake head that maps $\mathbf{z}_t$ to an 80-dimensional signed wake spectrum, both trained with small weights jointly with the representation and neither part of the latent-prediction loss. These heads are adopted from the observable-augmentation lineage (Fukami & Taira 2023, 2025). The wake target itself is a fixed descriptor chosen by design rather than tuned: it pairs a coarse signed spatial pooling of the wake region, which retains the layout of the alternating positive and negative vortices while discarding pixel-level detail, the spatial-pyramid construction of Lazebnik *et al.* (2006), with a radially-averaged power spectrum, the canonical isotropic scale descriptor of a turbulent field (Pope 2001) and an established compact image descriptor in machine learning (Durall *et al.* 2020), so it encodes both where the wake vortices sit and at which scales their energy resides. Pairing a localised spatial summary with a multiscale spectral one is the spatial-plus-scale representation formalised by the wavelet scattering transform (Bruna & Mallat 2013), which has been applied to turbulent flow fields precisely to recover geometry that the power spectrum alone cannot resolve (Skinner *et al.* 2025). We keep the descriptor compact, at 80 dimensions, so that it constrains the wake structure without over-parameterising the target on the available training encounters. They are also, however, the reason the present predictive encoder is not a clean isolation of the training objective: the reconstructive baseline below carries a lift head but not a wake head, and the two encoders differ in architecture as well. The headline comparison is therefore between encoder *families* as configured, and the controls of §4.2 separate the objective from the architecture and the wake supervision.

3.2. Matched-predictor closure protocol

The comparison rests on a single principle: every baseline is the same pipeline, and only the encoder differs. Each encoder produces a latent trajectory $\mathbf{z}_{1:T}$ from the vorticity fields, and an identical probe family maps a latent to a physical observable for evaluation (figure 3(b)). The probe family is held fixed across the families, and the downstream

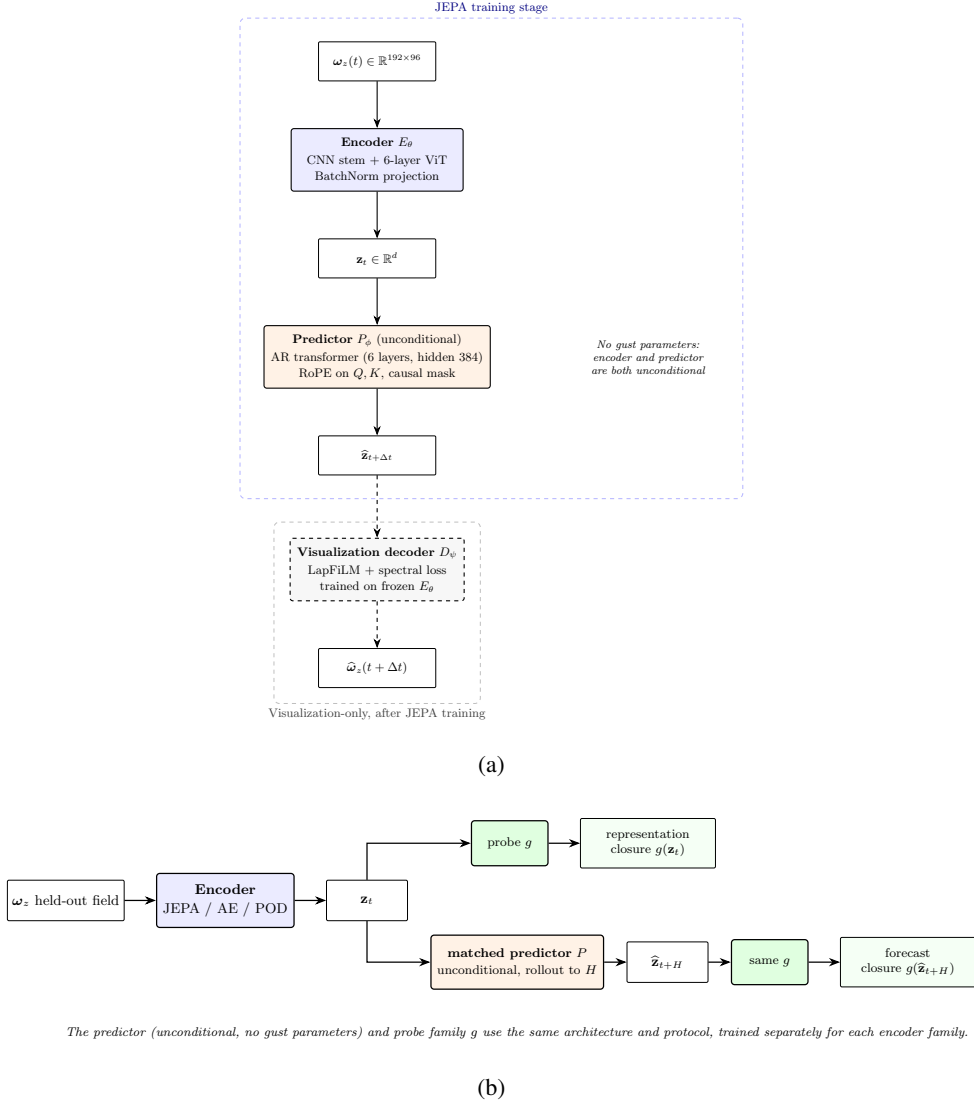


Figure 3. The predictive model and how it is evaluated, drawn schematically (the training predictor is the vector transformer of table 2). (a) The encoder maps the mid-plane vorticity field to a latent $\mathbf{z}_t$, and the predictor advances the latent from its own history; no gust parameters enter the encoder or the predictor. The two auxiliary observable heads (lift and wake) read $\mathbf{z}_t$ during training with small weights and are not part of the latent-prediction loss; they are omitted from the schematic for clarity. The visualisation decoder is trained only after the encoder is frozen and is used solely for the diagnostic decodes, never in the predictive loss. (b) Evaluation refits the same matched forecast operator and applies the same probe family g to every encoder family: a probe on the simulation-encoded latent measures representational closure, the headline, and the same probe on the rolled-out latent $\hat{\mathbf{z}}_{t+H}$ measures forecast closure, reported as secondary.

323 forecast operator is refitted identically on each frozen latent. For the forecast merit we
 324 use a single common operator, a residual U-Net, fitted with the same recipe and budget
 325 on every family, chosen because it is numerically stable across all the latent geometries;
 326 a transformer operator, though it is the flagship's training class, forecasts the predictive
 327 latent well but is unstable when refitted on latents outside its design, collapsing to a

negative merit on one reconstructive control (Appendix A), and so is not a usable common yardstick. The flagship’s own co-trained predictor, the as-built reduced-order model, is reported separately, and the sequential estimator of §3.3 uses a matched transformer on the flagship’s own trajectories, where it is stable. Because none of the reconstructive or objective-free baselines train a dynamics model at all, the common operator does not match any family’s training class more closely than another’s, so the merit column is a neutral read of forecastability rather than a home-class advantage. Every state is compared at the matched headline dimension $d = 32$; the dimension plateau at $\{16, 32, 64\}$ and the minimum-dimension panel of §4.6 establish that this choice is a robustness statement rather than a tuned operating point. This isolation is the only fair way to attribute representation quality to the encoder rather than to incidental differences in the readout. The reconstructive references use their best-documented configuration, with ReLU activations and group normalisation rather than the strict published variant (tanh, no normalisation), which we found gives a worse parametric probe; the comparison is therefore against a tuned baseline, not a hobbled one. The observable heads are pinned to the same current-frame form for every family, so no family gains an accidental advantage from a differently-shaped supervision signal.

The headline figure of merit is representational closure: a fixed linear probe is applied to the latent encoded directly from a held-out field, and we ask whether it recovers each of the five observables, the wake in particular. This isolates what the encoder carries, with no rollout. We report it as the held-out coefficient of determination R^2 on test_b and test_c, with the per-observable mean absolute error alongside, and bootstrap 95% confidence intervals over encounters. As a secondary, confirmatory measure we also report forecast closure: the predictor is rolled recursively from the pre-impact context to $H = 16$, and the same probe maps the predicted latent to each observable. A small forecast error means the encoder produces a latent whose dynamics, propagated by the predictor, retain the information the probe needs; a large error means either that the encoder discarded that information or that the predictor lost it during rollout, a distinction the drift diagnostic below resolves. The representational closure is in table 3, which reports the readability and the matched-predictor forecast merit side by side. We do not headline a forecast- R^2 -versus-horizon curve; the rollout is reported to show that the latent stays usable, not as a validated forecast.

3.3. The predict-correct estimator

At deployment the vorticity field is unavailable and the wall pressure is not, by itself, enough to fix the state: a single pressure frame recovers little of the wake through any of the latents (§4.3). The reason a single frame is not enough, and a window is, is a statement about the dynamics rather than about the number of sensors. The encounter lies on an attractor of low effective dimension, a limit cycle with a gust excursion and a relaxation, into which the encoded latent concentrates its variance (§4.6). By the delay-embedding theorem (Takens 1981; Sauer *et al.* 1991), a generic observation of such a system, read over a window of delays, reconstructs its state up to a diffeomorphism, and with K simultaneous taps over m delays the reconstruction is generic once the product mK exceeds twice the attractor dimension. Because that dimension is small, a handful of taps over a modest window suffices, and the delay window can substitute for sensor count, a trade we quantify in §4.3. A coordinate the instantaneous pressure misses, the wake circulation, is still reconstructed by the window provided it is dynamically coupled to what the pressure sees, which for the coupled vortex dynamics it is. The estimator must therefore be sequential, accumulating pressure innovations through the learned dynamics, which is the setting of data assimilation.

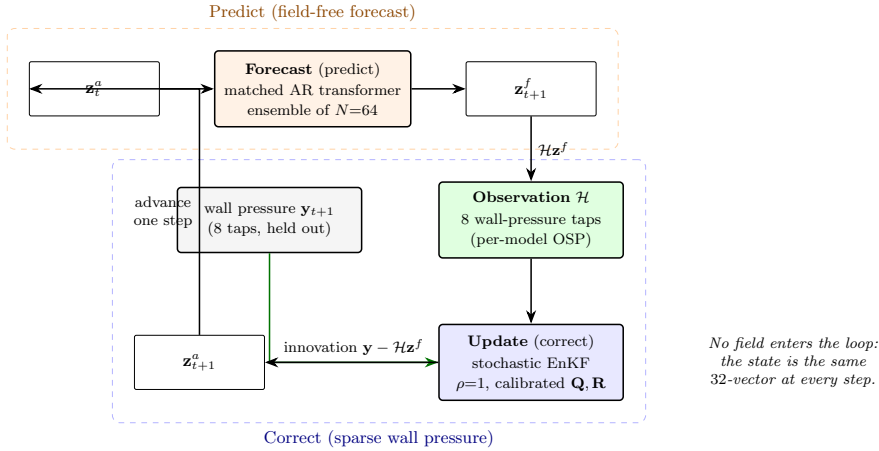


Figure 4. The leakage-free predict-correct estimator in the latent space. An ensemble of 64 members is advanced field free by the matched autoregressive transformer (predict), then corrected from 8 span averaged wall-pressure taps through the frozen observation operator $\mathcal{H}$ (correct); the innovation is the difference between the measured taps and the taps predicted from the forecast state. No vorticity field enters the loop and the lift and wake probes never enter the innovation, so the analysis closure is a genuine estimate. The state is the same 32 vector at every step.

We use an ensemble Kalman filter (Evensen 1994, 2003) in the latent space. An ensemble of $N = 64$ members is advanced one step at a time by the matched autoregressive transformer fitted once on the frozen training trajectories (§3.2), the forecast step, and corrected against $K = 8$ span-averaged wall-pressure taps through a frozen observation operator $h : \mathbf{z} \mapsto p_K$, the analysis step. The observation operator maps the latent to pressure only. The physical observables the paper cares about, the lift and the wake enstrophy, are read from the state by the same frozen probes used throughout and can never enter the innovation; this leakage guarantee is enforced by construction and by a unit test, so the reported analysis closure of the load and the wake is a genuine estimate, not a quantity the filter was given. The taps are placed by a model-conditioned criterion that selects, per family, the taps that best expose that family’s own latent. The filter is initialised, field free, from a windowed pressure-to-latent regressor, and its ensemble spread from that regressor’s residual covariance, so no encoded truth is used at inference. Process noise is calibrated from the predictor’s one-step residuals; a multiplicative inflation was selected on the validation split alone and then frozen, and it selected no inflation, because the filter is under-dispersed rather than over-dispersed and added inflation would hurt. All tuning is frozen before any test encounter is seen. Each encounter is assimilated over the window from 24 frames before impact to 48 frames after it. The baselines, all read through the same probes, are the open-loop rollout with no correction, a sliding-window pressure-only regression with no dynamics, and a model-free parameter floor in which the gust parameters are given, used diagnostically only, the model never seeing them.

The mixed timescales of this estimator, a transition map trained on sequences and an emission map trained per instant, mirror the structure of the training objective in §3.1: a dynamics equation is a statement about trajectories and an observation equation is a statement about single instants. The combined loss is a jointly trained state-space realisation, and the filter is the same decomposition read at inference.

3.4. *Diagnostics, controls, and uncertainty*

Two diagnostics turn the closure result into a mechanism. The first is the latent drift, which addresses the fact that a predictor with low one-step error need not produce a useful long rollout, because each prediction is fed back as the next input and small geometric errors accumulate. We quantify how far a rollout has left the training manifold by the Mahalanobis distance of the rolled-out latent at horizon H , measured against the distribution of simulation-encoded latents for the same encoder, normalised by the Mahalanobis distance of the simulation-encoded latent itself. A ratio near one means the rollout stays within the natural spread of the training latents; a large ratio means it has gone out of distribution, so that the probe is then queried outside its support.

The second diagnostic is a parameter-only floor, which carries particular force here because the model never sees the gust parameters: it bounds what the parameters alone could explain, so that any closure above it is genuinely supplied by the latent state. We compute it with a kernel-ridge regressor that maps $\mathbf{c} = (G, D, Y)$ directly to each observable at $H = 16$, with no latent input, fit on training and evaluated on the held-out splits. On wake enstrophy this floor is low, so the latent’s wake closure is not something the parameters could have supplied. At the impact frame the same parameter floor is high, because the just-released gust nearly determines the instantaneous state; the gap opens as the wake evolves (§4.1).

Because the predictive encoder differs from the reconstructive baseline in objective, architecture, and auxiliary supervision at once, the comparison is accompanied by controls that vary one factor at a time: a crossing of objective with architecture at matched auxiliary heads, an ablation of each auxiliary head, and a conditioned predictor variant that adds $\mathbf{c}$ (§4.2). Every held-out R^2 and mean absolute error reported below carries three independent uncertainty signals, following the reporting protocol fixed with the data split: a 2000-resample bootstrap over the held-out encounters, the standard deviation across three encoder retrains from independent seeds, and a five-fold cross-validation over cases on the probe read-out. Their full specification is in Appendix A.

4. **Results**

We report the controlled representational comparisons (§4.1 to §4.3) on the in-distribution test_b encounters and hold the out-of-distribution test_c set in reserve, so that the $|G| = 4$ extrapolation enters only as a final reported number and never as a selection signal. The estimator, the recovery, and the operating envelope (§4.4–§4.5) are characterised with a frozen filter across all encounters, train, validation, test_b and the test_c boundary alike, and the per-encounter paired tests on test_b and test_c are given in Appendix A. Every quantity quoted below is a held-out value at the readout horizon unless stated otherwise.

4.1. *What the coefficient state carries, and what supplies it*

The comparison between reduced states is meaningful only if the wake is a discriminating endpoint, and it is. At the readout horizon the release parameters no longer fix the wake enstrophy, while they still fix the forces, so the wake is where a representation that has captured the gust-vortex physics parts company from one that has captured only its force signature. Table 3 reports what each pooled coefficient state carries, read by a fixed linear probe from a held-out test_b field with no rollout. The wake is read cleanly by every state that carries the wake-supervision head, with a held-out R^2 between 0.750 and 0.792, and it collapses for every state that does not, falling to 0.160 for the predictive latent trained without the head. Instantaneous readability of the wake is therefore supplied by the observable supervision, not by any training objective.

Table 3. What each pooled coefficient state carries, at matched dimension $d = 32$ on the 42 test_b encounters. Held-out wake-ensrophy readability (fixed linear probe, no rollout), matched-predictor forecast merit (mean over the five observables at horizon sixteen), field-reconstruction error (VRMSE at horizon sixteen), and decode similarity (SSIM). The controlled matrix (top block) moves one training ingredient at a time; the reference recipes (bottom block) use their own architectures. Wake readability separates on the wake-supervision head, not on the training objective.

State	Objective	Wake head	wake R^2	merit	field VRMSE	SSIM
JEPA (wake)	predictive	yes	0.751	0.591	0.962	0.779
supervised only	none	yes	0.792	0.637	0.992	0.774
AE (wake)	reconstruction	yes	0.750	0.548	0.976	0.776
JEPA (no wake)	predictive	no	0.160	0.471	0.983	0.767
AE (no wake)	reconstruction	no	0.456	0.432	0.976	0.773
anti-collapse AE	reconstruction	no	0.333	0.197	0.949	0.782
β -VAE	reconstruction	yes	0.728	0.356	1.007	0.724
Fukami	reconstruction	no	-0.094	-0.200	1.016	0.708
Fukami (wake)	reconstruction	yes	0.432	0.203	1.026	0.705
POD	linear	no	0.186	0.290	0.930	0.775

The objective-free supervised encoder makes this concrete. Trained with the wake head but no predictive or reconstructive objective at all, it reads the wake at least as well as the full predictive latent (0.792 against 0.751) and matches it, indeed slightly leads it, on the merit a common matched operator extracts by forecast (0.637 against 0.591). On this tier, then, we do not attribute the readability or the common-operator merit to the predictive objective over the supervised control: the paired readability gap is 0.041 in the wake with a confidence interval $[-0.047, 0.162]$ that includes zero, and under the common operator the two supervised states sit together and above the reconstructive and anti-collapse recipes ($0.637 > 0.591 > 0.548 > 0.197$). What the predictive objective adds shows not in this common-operator read but in the as-built model it produces: the flagship's own co-trained predictor forecasts the five observables at 0.755 at the same horizon, well above what the common operator extracts from any latent, which is the reduced-order model a deployment would actually run. Across three independent retrains the predictive state's common-operator merit spreads by 0.098, tighter than the objective-free control's 0.163 but not the sharpest of the pooled families, so at this sample size we read seed stability as a caution rather than a discriminator (§5.4). We state the representational result as the supervision's, and turn to what the predictive objective does add, which is dynamical.

One inversion in the table fixes the coordinate-level reading. The predictive latent trained without the wake head carries the highest single-observable lift forecast of any state (0.691) yet the lowest wake readability (0.160), because the forces are held redundantly across the latent while the wake is not, the distributed-code signature we return to in §4.6. Among the reference recipes the lift-augmented reconstructive autoencoder reads the wake only when it carries the wake head (0.432 against -0.094 without it), and the linear basis does not reach the wake at all (0.186), so no linear projection recovers the enstrophy that a supervised nonlinear state does.

4.2. Why the state stays usable under rollout

A state that reads the wake at a single instant need not survive a rollout, because a probe attached to a drifted latent is queried outside its support. We ask where each rolled state goes relative to the training distribution by decomposing its departure onto the principal and the near-null directions of the encoded covariance (table 4). The reconstruction carrying only an anti-collapse term is near-isotropic, its covariance condition number close to unity

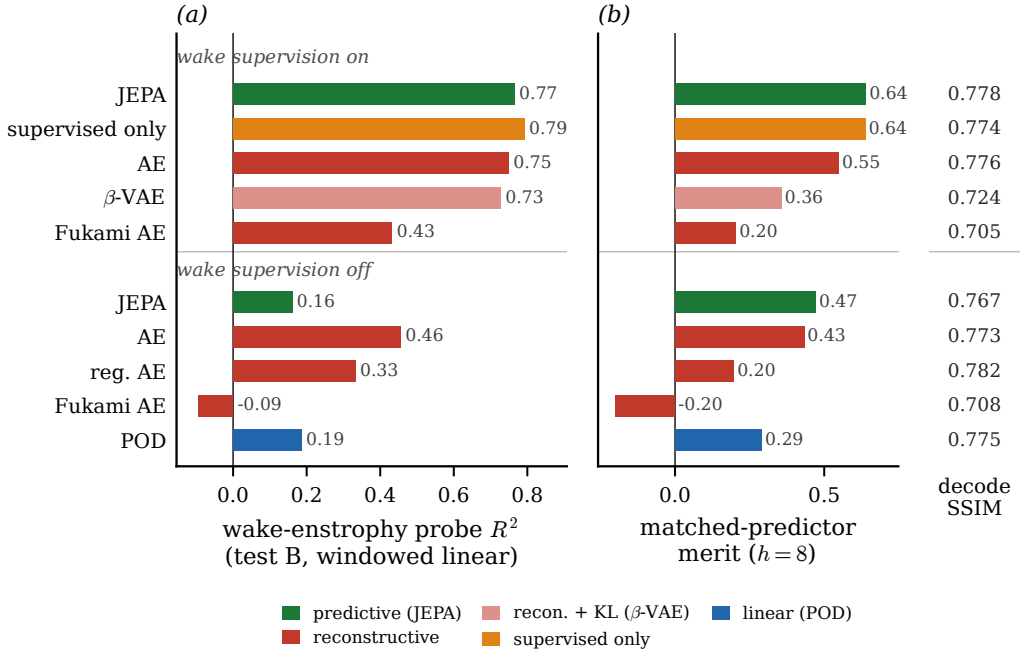


Figure 5. The attribution result of table 3, at a glance. (a) Held-out wake-entrophy readability and (b) matched-predictor merit, with the models grouped by whether they carry the wake-supervision head; every wake-headed state reads the wake and every bare state collapses, whatever the training objective. The right-hand column lists the decode similarity, which barely separates the families.

(11), and its rollout leaks a tenth of its departure into the near-null directions (0.100). The supervised states, predictive and objective-free alike, are strongly anisotropic, their variance concentrated in a low-rank subspace (condition numbers 451 and 1041), and their rollouts keep their departures inside it (0.014 and 0.029, against the 0.25 of an isotropic null). Supervision, not the anti-collapse term alone, builds the protected subspace: it pins the encoded variance onto the observable-aligned directions, and the forward model, having only ever seen states there, moves within it. The anti-collapse term on its own drives the latent toward isotropy, as an independent adaptation of the same regulariser to biosignals also reports (Broustail *et al.* 2026), and it is the supervision that supplies the geometry the rollout needs.

What the predictive objective adds is long-horizon stability, and it is the multi-step form of the objective that adds it. Trained with a single-step rather than a multi-step rollout, the same predictive latent forecasts the five observables less well at every horizon, and the margin widens with the rollout: by sixteen steps the single-step latent’s merit has fallen to 0.237 while the multi-step latent holds 0.488, with markedly lower on-manifold drift (0.518 against 0.733 at horizon sixteen). The single-step latent is marginally better on the load alone a few steps out (0.685 against 0.664 in C_L closure at eight steps), a single-observable exception to a forecast advantage that otherwise grows with horizon. The objective-free supervised state, which has the readability and the protected subspace but no predictive training, is the weakest field forecaster of the supervised group. The contribution of the predictive objective is therefore forward rather than instantaneous: it keeps the rolled state both in distribution and near the true trajectory, which is exactly what a sequential estimator queries.

Table 4. Where each rolled state goes relative to the training distribution, on test_b at rollout horizon sixteen. The latent covariance condition number measures anisotropy, and the near-null departure fraction is the share of the rollout departure that falls into the bottom-quartile directions of the encoded covariance. An isotropic null would leak 0.25. The anti-collapse reconstruction is near-isotropic and leaks; the supervised states are anisotropic and stay in their low-rank subspace.

State	covariance condition number	near-null departure fraction
anti-collapse AE	11	0.100
JEPA (wake)	451	0.014
supervised only	1041	0.029

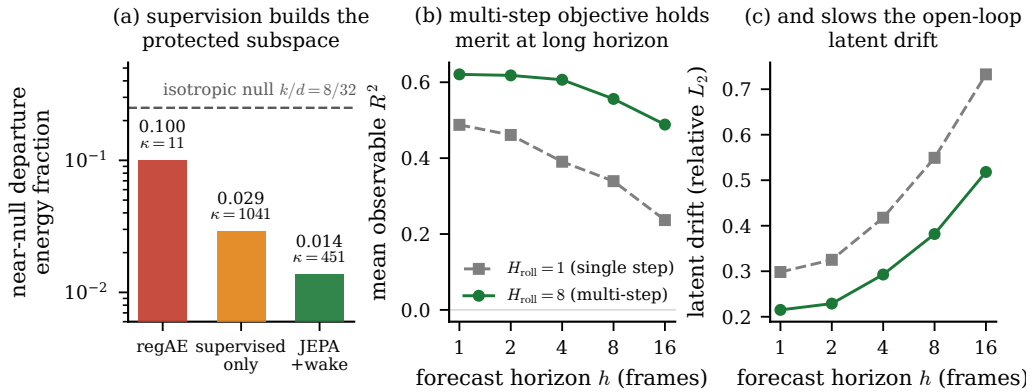


Figure 6. The rollout mechanism and what the multi-step objective adds. (a) Near-null departure fraction by family against the isotropic null, annotated with each latent covariance condition number: supervision builds the protected subspace. (b) Forecast merit and (c) rollout drift against horizon for the single-step and multi-step objectives, tied at one step and separated by sixteen.

4.3. What the wall can see

At deployment the state must be read from the wall, so we ask how much of each latent a sparse set of pressure taps recovers over a window ending at the readout instant (table 5). Two findings organise the answer. First, raw-variance recoverability and physics recoverability disagree: the reconstruction latent is the most recoverable in state variance (0.921) yet carries the least wall-recoverable physics (0.470 in observable closure), while the predictive latent carries the most physics (0.589) from a less recoverable state (0.659 ± 0.009 over three seeds). The wall aligns with many of the reconstruction latent's directions, but they are the low-variance ones that do not carry the observables; it aligns with fewer of the predictive latent's directions, but those are the ones that hold the state. Second, perturbing the predictive state along its principal directions and measuring the signal that reaches the pressure head shows the wall reading the force and wake-entropy directions and blind to the highest-variance wake-circulation direction, a ranking unchanged under a fixed-amplitude perturbation and so a property of the learned dynamics rather than of the normalisation. The wall, in short, reads the force well and the wake circulation not at all, which is why the state must be estimated sequentially rather than inverted from one frame.

The margin of the coefficient state over a raw high-dimensional field latent is real but modest. Its recoverability advantage is 0.120 in R^2 , with a confidence interval $[0.096, 0.145]$ that clears zero but not the pre-registered strong-effect bar of 0.2. We therefore do not claim a strong-form estimability gate; the defensible statement is the clean physics-versus-variance ordering above.

Table 5. What eight wall-pressure taps recover of each family’s coefficient state, on test_b, with the recovery estimator selected by cross-validation per family and the observables read through the frozen probes. Raw-variance recoverability and physics recoverability disagree: the reconstruction latent is the most recoverable in state variance but the least in physics, while the predictive latent carries the most wall-recoverable physics from a less recoverable state.

State	recovery estimator	state R^2	observable R^2
predictive (JEPA)	LSTM	0.659	0.589
reconstructive (Fukami)	LSTM	0.921	0.470
linear (POD)	MLP	0.561	0.534

4.3.1. Sensors traded for delays

That the wall is blind to the wake circulation at a single instant is a statement about one frame, not about the sensor count, and the delay-embedding reading of §3.3 makes a testable prediction: the circulation the instantaneous pressure misses should become recoverable as the pressure is read over a longer window. It does. Holding the tap count at eight and lengthening the window, the recovery of the wall-blind wake circulation rises from near its single-frame value to 0.33 at a thirty-frame window, a gain of 0.224 with a case-clustered interval $[0.136, 0.343]$ that excludes zero, while the force, already visible at one instant, barely moves. The delays recover the coordinate the sensors cannot.

Varying the tap count and the window together traces a spatial-for-temporal trade (figure 7). A single tap read over a thirty-frame window recovers the coefficient state as well as eight taps read at one instant, and intermediate combinations of tap count and window meet in between, because the encounter is low-dimensional and a state that filled its ambient dimension would not admit the trade. The eight-tap, thirty-frame recovery reconciles exactly with the static recovery above (0.656), and the delay stride sits at the cache cadence, close to the first mutual-information minimum of the tap signals at 29 frames.

The grid uses a shared target-blind tap array so that the trade is free of any model-conditioned placement; rerunning its eight-tap, thirty-frame cell on the model-conditioned taps of table 5 reproduces that table’s recovery exactly (0.656 against 0.659), so the two analyses meet where their sensing coincides. This trade belongs to the static delay-coordinate reconstruction, and it does not transfer to the frozen sequential filter, which needs its eight taps. Rerunning the filter with only its own leading taps, and changing nothing else, drives the analysis lift below the skill of the encounter mean at every gust strength the eight-tap filter tracks: at $|G| = 1$ the single-tap analysis is worse than the eight-tap filter by a paired margin whose interval $[-107.96, -4.78]$ excludes zero, and the loss stays significant at $|G| = 1.5$ and $|G| = 2$. The rank of the innovation, with the process and observation noise held at their calibrated values, is what the filter cannot spare: its sensor budget is a calibration constraint, not a delay-embedding one, which points at the noise model of §5.4 rather than at fewer sensors.

4.4. Tracking the encounter from the wall

We now run the estimator. It is initialised from wall pressure alone before impact and corrected at every frame through the encounter, and it never sees the vorticity field. Figure 8 follows representative held-out encounters, the true lift and wake enstrophy against the open-loop rollout with no correction, the sliding-window pressure-only regression with no dynamics, and the filter analysis. The open-loop rollout tracks the pre-impact baseline and then loses the encounter, as a forecast with no measurement must; the pressure-only

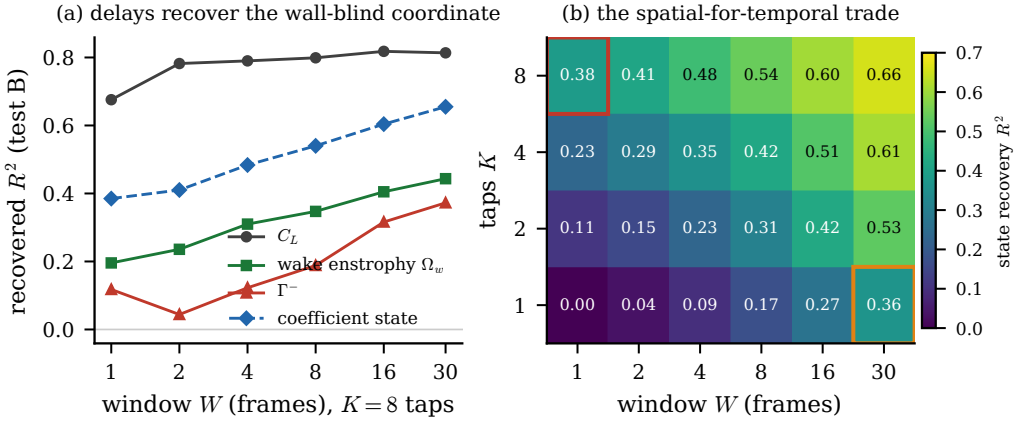


Figure 7. Sensors traded for delays. (a) Recovery of each observable from eight wall-pressure taps against the length of the pressure window (test_b); the wall-blind wake circulation rises with the window while the force, visible at one instant, does not. (b) Coefficient-state recovery over the tap-count by window-length grid, with the single-tap, thirty-frame cell and the reference eight-tap, single-frame cell outlined: a long window with few taps matches a short window with many.

regression follows the force while its window is representative and goes stale through the transient; the filter follows the load through impact and into relaxation. This realises, on the wake-supervised coefficient state, the online-estimation pathway the extreme-aerodynamic control literature has named but not closed (Fukami *et al.* 2024; Mousavi & Eldredge 2025; Eldredge & Mousavi 2025): the state a real-time controller would need is recovered from sparse wall pressure and advanced by the learned dynamics, without ever measuring the field.

Two honest boundaries frame what this demonstration does and does not establish, and we state them before the envelope. First, tracking the load inside the training range is not unique to this state. Running the identical frozen filter on every family (table 6), the nonlinear latents all track the lift for $|G|$ between one and two, and the wake-supervised reconstructive control is the best of them at the mildest gusts (0.86 at $|G| = 1$ against 0.74 for the predictive state); only the linear basis fails outright. What no static approach achieves is the strong-gust range: a windowed regression from the same taps straight to the lift, with no latent at all, fades to 0.08 by a gust ratio of three, and even the full wall array recovers only 0.14 at the boundary, so the sequential accumulation of pressure through a learned nonlinear state, rather than any particular training objective, is what carries the load into the strong-gust regime. Where the objectives separate is at that boundary and in the filter's statistical health: the predictive state alone combines the highest boundary closure (0.84 against 0.66 to 0.85 for the alternatives), the lowest divergence rate at every gust strength, and the least-degraded wake readout. Second, no family's filter tracks the wake enstrophy: the analysis wake closure is below the mean predictor for every state, least badly for the predictive one, so the wake content of the encounter is established at the representational tier (§4.1) and is not yet recoverable through the assimilation. The filter is a load estimator whose state carries wake information; it is not a wake estimator, and we say so plainly.

How far into the gust the load recovery holds is the operating envelope of §4.5.

Table 6. The frozen filter run identically on every family: median analysis lift closure by gust strength, with the divergence rate at the $|G| = 4$ boundary, and the direct no-latent baseline (windowed regression from the same eight taps straight to the lift) beneath. Load-tracking inside the training range is a property of the nonlinear latents generally, not of any one objective; the families separate at the strongest gusts, in calibration, and in what else the tracked state carries (§4.4). The linear basis and the no-latent regression both fail beyond $|G| = 2$.

state	$ G = 1$	1.5	2	3	4	div. rate at 4
JEPA (wake)	0.74	0.83	0.73	0.63	0.84	0.72
AE (wake)	0.86	0.86	0.81	0.49	0.66	0.93
Fukami	0.79	0.87	0.78	0.68	0.76	0.97
Fukami (wake)	-0.39	0.72	0.68	0.77	0.85	0.95
POD	0.53	0.52	-0.33	-0.27	0.25	0.93
direct regression (no latent)	0.41	0.29	0.21	0.08	0.08	—

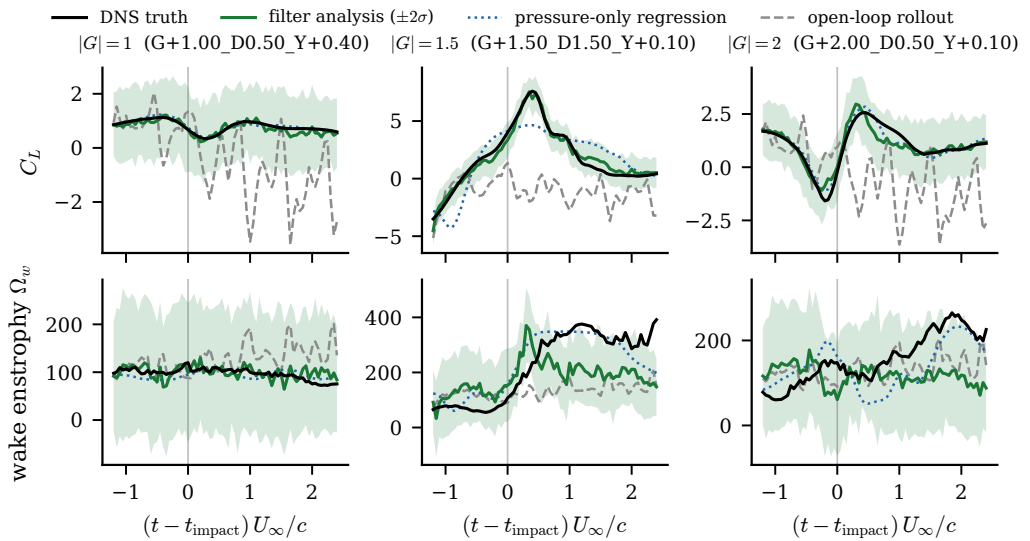


Figure 8. The estimator tracks the encounter from wall pressure alone. Columns are representative held-out encounters at $|G| = 1, 1.5$ and 2 ; rows are the lift coefficient and the wake entropy. Each panel shows the DNS truth, the filter analysis with its $\pm 2\sigma$ ensemble band, the pressure-only regression, and the open-loop rollout; the vertical line marks impact. The filter follows the load through impact and relaxation where the open-loop forecast loses it and the pressure-only regression goes stale.

4.5. The gust-intensity operating envelope

A reduced state earns its place at deployment only if it can be followed from the sensors that are present, and the wall pressure a small vehicle can carry does not, at a single instant, fix the wake it leaves behind (§4.3). The question is therefore not whether the state is readable but how far into the gust an estimator that sees only the wall can track it. The sequential filter reaches much further than the single-frame inverse. While the encounter stays mild, the static single-frame recovery of the lift holds, with a held-out R^2 of 0.73 at $|G| = 1$ and 0.65 at $|G| = 1.5$; as the gust strengthens it collapses, falling below the skill of predicting the encounter mean by a gust ratio of three (-0.46) and remaining there at the $|G| = 4$ boundary (-0.33), because a single pressure frame cannot see the reorganised wake a strong gust deposits. Reading the same taps through the learned dynamics, the filter still follows the load across those encounters, from 0.74 at $|G| = 1$ to 0.63 at a ratio of three and 0.84 at the boundary (table 7, figure 9). The open-loop forecast never tracks the load at

Table 7. The operating envelope, by gust strength, for the predictive coefficient state under the frozen filter. Median analysis lift closure and divergence rate of the filter against the static single-frame recovery, over all encounters (train, validation, test_b, and the $|G| = 4$ test_c boundary), stratified by $|G|$. The $|G| \leq 0.5$ closure is near zero because the true lift is near-constant there and the R^2 denominator vanishes, not because the filter fails; the per-encounter paired tests on test_b and test_c are in Appendix A.

$ G $	filter C_L	R^2	filter divergence rate	static recovery C_L	R^2
≤ 0.5	0.25		0.00	0.34	
1	0.74		0.00	0.73	
1.5	0.83		0.09	0.65	
2	0.73		0.33	0.24	
3	0.63		0.68	-0.46	
4	0.84		0.72	-0.33	

any gust strength, so it is the correction step, not the forward model, that recovers the state, and it does so from pressure the visibility analysis found blind to the wake circulation: the sequential dynamics carry the part of the state the wall cannot see at an instant.

What weakens with gust strength is not the tracking but the filter's own account of its uncertainty. The divergence rate climbs from none at $|G| = 1$ to four encounters in five at the boundary, and the mean normalised innovation squared from near unity in the undisturbed limit (0.6) to nearly twenty at $|G| = 4$ (15.2), crossing the half-divergence mark near a ratio of 3.0 for the compact cores and near 3.0 for the widest. The ensemble collapses while the analysis mean still follows the load, so the filter is under-dispersed rather than mistracking. We read this as the signature of a missing noise model: the innovations are correlated in time rather than white, and the remedy is a calibrated process and observation covariance rather than the covariance inflation we deliberately withheld (§5.4).

The envelope is not symmetric in the sign of the gust, and the asymmetry survives in the linear basis, so it belongs to the flow rather than to the representation (figure 9(b)). The physically positive gusts, which roll up the larger and longer-lived leading-edge vortex (§4.6), leave the wider usable envelope. Sampling the $|G| = 4$ boundary at both signs (§2.2) is what lets this asymmetry be measured rather than assumed.

The boundary sits at a gust ratio of three, inside the training range, so it marks a limit of what the mid-plane pressure can observe and the filter assimilate, not a gap in training coverage. Read through delay coordinates the two limits are one. The filter follows the load where the single frame fails because it reconstructs the state from a window of pressure rather than an instant (§3.3), and it loses its calibration where the strengthening gust drives the interaction three-dimensional and the encounter's effective dimension outgrows what a fixed tap count and delay window can embed (§4.3). Only at $|G| = 4$ does the mid-plane representation meet the genuine three-dimensional observability boundary the field itself reaches (§2.1), now seen from the filter. The envelope is therefore two claims with different owners: its extension beyond every static inverse belongs to the sequential estimator on a nonlinear latent, a property the family comparison of §4.4 shows is shared; its width at the strongest gusts, its calibration margin, and the wake content of the state being tracked belong to the wake-supervised predictive state. This envelope, not the single-instant readability, is what a wall-estimable reduced state is for.

4.6. The physics the latent state holds

The estimation results rest on a representation, and this subsection reads what the representation holds in measures that do not depend on the filter. The predictive latent carries the shedding clock. Fitting a linear operator to the encoded limit-cycle trajectory

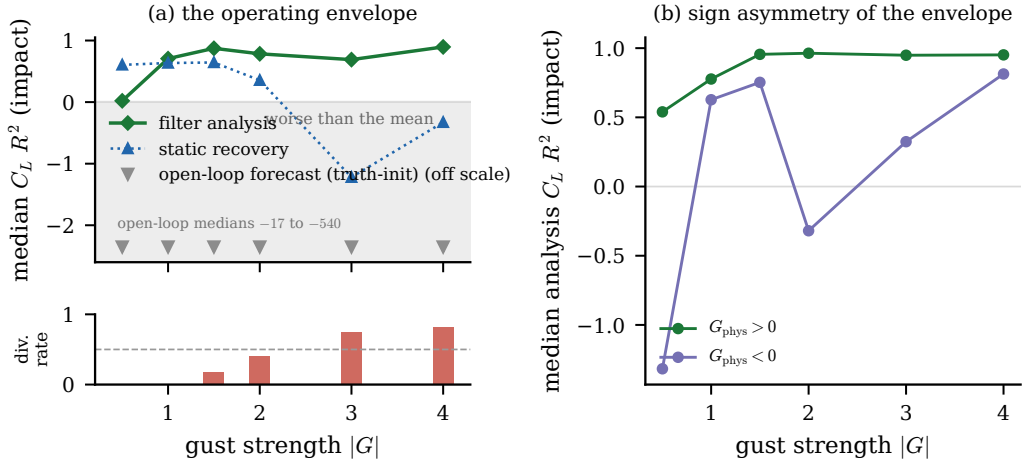


Figure 9. The gust-intensity operating envelope. (a) Median analysis lift closure (R^2 , impact window) against gust strength for the frozen filter, the static single-frame recovery, and the truth-initialised open-loop forecast, with the filter divergence rate as the strip beneath; the shaded band marks closure worse than predicting the encounter mean, and the open-loop forecast lies off the scale below it. (b) The same filter closure resolved by the physical sign of the gust, showing the wider usable envelope of the positive gusts.

by dynamic mode decomposition places a marginally stable oscillatory mode at Strouhal number 0.666, against the measured shedding line, with modulus 0.993 just below unity. The Fukami-lineage autoencoders lose that clock: their leading latent mode is heavily damped and off-frequency (0.182 at modulus 0.590 without the wake head, 0.147 at 0.294 with it). The structured states keep it, whether their structure comes from prediction, from supervision, from anti-collapse, or from a linear basis: the objective-free supervised latent, the anti-collapse reconstruction, and the proper orthogonal basis all place the shedding mode near the measured frequency on a near-unit-modulus orbit (0.654, 0.676 and 0.673 at moduli 0.989, 0.995 and 0.996). It is the Fukami-lineage reconstruction, not reconstruction in general, that fails to hold the shedding spectrum.

The wake is a collective code rather than a labelled coordinate. The gap between the wake readability of the full predictive latent and that of its single most informative coordinate is 0.51, and 0.53 for the objective-free supervised state, against -0.03 and 0.04 for the reconstructive and linear baselines, so no single predictive coordinate carries the wake and the information is spread across many. The dimension at which each family first reaches a wake R^2 of one half makes the same point from the other side: the predictive and the wake-headed reconstructive states need eight latent dimensions (8 each), while the linear basis never reaches it at any dimension we tested. The headline dimension of thirty-two sits on a plateau rather than a cliff, the wake readability moving only 0.053 across sixteen, thirty-two and sixty-four (0.751 at thirty-two), so the choice of dimension is a robustness statement and not a tuned operating point.

The parametric manifold coheres with the release parameters where the training data resolve them. An impact-frame probe of the predictive latent recovers the gust strength and the core diameter well (0.88 and 0.73 on test_b), which is the atlas of figure 10 coloured by parameter. The wall-normal offset, unreadable at the linear-probe level in earlier work, becomes weakly readable here (0.29 linearly and 0.79 with a nonlinear probe), plausibly because the additional cases sample it more densely near the training centre; we note this rather than lean on it, since the offset remains the least resolved parameter (§2.2).

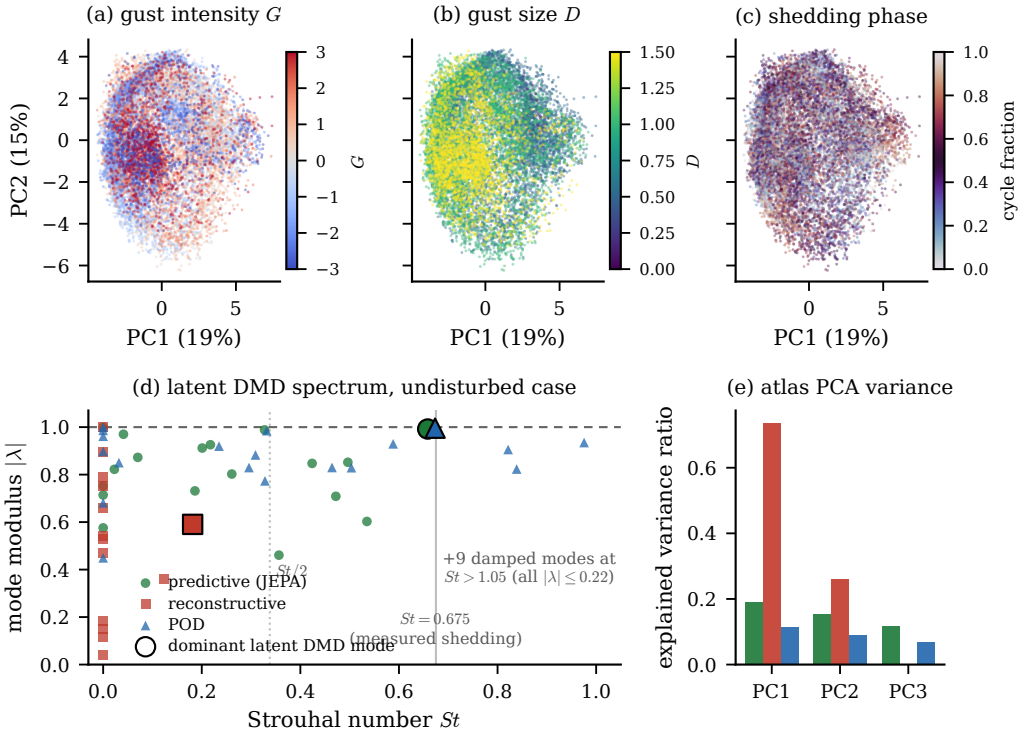


Figure 10. The physics the latent state holds. Top: two-dimensional projections of the predictive coefficient latent coloured by gust strength, core diameter, and shedding phase, showing a parametric manifold that coheres with the release parameters. Bottom: the latent dynamic-mode-decomposition spectrum, modulus against Strouhal number, by family; the predictive and linear latents place a marginally stable mode at the measured shedding frequency while the Fukami-lineage reconstruction is damped and off-frequency.

A spatially resolved latent decodes the vorticity field more sharply than this coefficient state, but it is of far higher dimension than the wall measurement and so is not recoverable from sparse pressure. That decodability-versus-estimability trade is the reason the coefficient state, not the field-resolved latent, is the object this paper estimates; we take it up in §5.3 and quantify its cost in Appendix C.

5. Discussion

5.1. What a wall-estimable state must contain

The central result is a design rule for a reduced state that a wall-pressure estimator can track. Such a state must carry the wake observables that govern the post-impact transient, must be organised so that a forward model can advance it without leaving its own support, and must be recoverable from the surface pressure that is the only measurement a vehicle carries. These three requirements are supplied by different ingredients, and separating them is the paper's contribution. Observable supervision on the wake supplies the first, the instantaneous readability, and, less obviously, much of the second: it concentrates the encoded variance into a low-rank subspace aligned with the observables, and the forward rollout, trained and living in that subspace, stays in distribution, where an anti-collapse regulariser applied to a bare reconstruction produces a near-isotropic latent with no protected subspace and a rollout that leaks into the directions the training data never

constrained (§4.2). The mechanism is geometric, and it refines our earlier reading, which credited the anti-collapse term: a reconstruction objective fixes the latent only up to a diffeomorphism (Kelshaw & Magri 2024), the anti-collapse term on its own drives the latent toward isotropy (Broustail *et al.* 2026), and it is the supervision that pins the variance onto the observable-aligned directions the rollout respects. The multi-step predictive objective supplies the rest of the second requirement, the long-horizon stability that a single-step objective and an objective-free supervised encoder both lack, and, across retrains, the reliability of the forecast itself (§5.4). The third requirement, wall recoverability, is where energy and information part company: the reconstruction latent is the most recoverable in raw variance but the least in physics, while the predictive latent carries the most wall-recoverable physics (§4.3), so the coefficient state a controller would want is not the one a reconstruction objective produces.

The wake, and not the forces, is where these requirements separate the families, and the coordinate-level reading of §4.6 explains why. The forces are carried redundantly in every family, so an energy or reconstruction objective retains the force signature of an encounter without retaining the distributed combination of coordinates that carries the vorticity redistribution producing it. The gust parameters alone predict the forces well and fail on the wake at the readout horizon, so a state can look accurate on the load while carrying none of the physics the load is built from, which is exactly the failure mode the estimation results expose. The linear basis, for its part, remains genuinely useful: it is competitive on the body forces, amplitude-accurate in reconstruction, and it holds the shedding clock as well as the predictive latent (§4.6); the result is not that POD is obsolete but that no linear projection reaches the wake at any dimension we tested.

5.2. The estimation demonstration and its limits

The filter turns the design rule into a working estimator: from eight wall-pressure taps and no field measurement it tracks the load through the encounter and extends the usable gust-intensity range beyond every static inverse, realising the online-estimation pathway the extreme-aerodynamic control literature has named (Fukami *et al.* 2024; Mousavi & Eldredge 2025; Eldredge & Mousavi 2025). We are deliberate about its limits, and the first is one the family comparison forces: the in-range load-tracking is a property of sequential estimation on a nonlinear latent generally, not of this state, so the paper does not rest the representational claim on it. What the predictive state contributes to the estimator is the strong-gust boundary, the calibration margin, and a tracked state whose coordinates carry the wake; what no state yet contributes is a filtered wake estimate, since the analysis wake closure is below the mean predictor for every family. Its recoverability advantage over a raw high-dimensional field latent is real but modest (0.120 in held-out R^2 with a case-clustered interval [0.096, 0.145] over the train and test_b recovery rows, test_c untouched; §4.3), so the case for the coefficient state rests on the physics-recovery ordering and the envelope, not on a large recoverability margin. Its point tracking is good but its uncertainty is under-dispersed at the strongest and widest gusts, where the innovations are temporally correlated and the ensemble collapses. The visibility analysis suggests the mechanism: the forward model has large gain along the data-unconstrained near-null directions it never learned to damp, and those inject spread-consuming noise during assimilation. This points to a calibrated process and observation noise model, rather than covariance inflation, as the outstanding requirement, and we report the filter as a feasibility demonstration with a characterised soft spot, not as a deployed estimator.

The delay-embedding view names the deployment knob and its limit. Because the encounter is low-dimensional, the state can be tracked from few taps read over a short window, which is the sensing budget a small vehicle could carry; the trade between tap

count and window length is explicit for the static reconstruction (§4.3.1), so a sensor-poor platform can in principle compensate with a longer window within the encounter’s own timescale. The frozen filter does not inherit that trade, because its innovation rank, not the embedding, is what binds at reduced tap count, and the two statements together sharpen the same conclusion: the noise model, not the sensor count, is the binding constraint. The limit that no budget escapes is the growth of the encounter’s effective dimension with gust strength. Read through delay coordinates the encounter is an intermittently forced low-dimensional system (Brunton *et al.* 2017), the baseline shedding carrying the delay-coordinate dynamics and the vortex impact acting as a forcing whose amplitude is the gust ratio, which is consistent with the envelope being organised by $|G|$: no fixed sensing budget embeds the strongest, most three-dimensional encounters, which is why the envelope closes where it does (§4.5).

5.3. *Decodability versus estimability*

A reduced state can be optimised for two deployments that pull apart. A spatially resolved latent decodes the vorticity field more sharply than the coefficient state, but it is of far higher dimension than the wall measurement and so is not recoverable from a sparse set of pressure taps. The coefficient state decodes the field less sharply and is the object a wall-pressure filter can track. We therefore separate the two roles: the coefficient state is the estimable, forward-usable object this paper is about, and the field-resolved latent is a decodability instrument rather than an estimation target. Pooling the encoder onto the coefficient state costs field fidelity across every family, by -0.093 in decode similarity and -0.095 in matched-predictor merit for the predictive latent (Appendix C), and it gains nothing in wall recoverability that the coefficient state does not already have. The state a controller would carry is therefore not the one a field-reconstruction objective produces.

5.4. *Limitations and outlook*

Five limitations bound the claims. First, the attribution of the instantaneous results is honest but unglamorous: the wake readability is supplied by the observable supervision and its tie between the predictive and reconstructive wake-headed states is seed-robust (0.727 ± 0.021 against 0.760 ± 0.019), and the single-seed tie in matched-predictor merit between the full predictive state and the objective-free supervised control does not replicate across retrains. Read through a common matched operator, the merit does not cleanly separate the families: over three seeds the supervised control is seed-fragile (spread 0.163 , with one healthy retrain in three collapsing), the predictive latent is tighter but not the sharpest (± 0.098), and its seed-mean sits level with the matched reconstructive control (0.551 against 0.563), because that common operator is off-class for the transformer-trained state and understates it. The defensible dynamical claim rests instead on the as-built model and its rollout: the flagship’s own predictor forecasts markedly better than any common operator extracts from any latent, and the multi-step objective buys long-horizon stability and lower drift, not a larger instantaneous readability margin. Second, the wall-normal offset Y remains the least resolved release parameter because the training mass concentrates near $Y \approx 0$; the additional cases make it weakly readable where it was unreadable before (§4.6), and we note this without leaning on it. Third, the $|G| = 4$ extrapolation degrades for a physical reason: the encoder observes a single mid-plane vorticity slice whose information no longer determines the three-dimensional dynamics there (§2.1), so test_c marks the observability boundary of that representation, not a parametric-generalisation claim, and the filter meets the same boundary from the sensing side (§4.5). Fourth, the visualisation decoder is a diagnostic on the frozen encoder, never part of the predictive loss, and the pooled coefficient state trades decode sharpness for estimability by construction (§5.3). Fifth, and foremost

for what comes next, the filter’s calibration is the priority open problem: its uncertainty is under-dispersed at the strongest gusts, and the natural next step is a learned or physically motivated process and observation covariance, ahead of any closed-loop use.

These limitations come into sharpest focus against the use the architecture is meant for. The predictor is, by construction, a latent dynamics function of the kind an estimator or controller requires, the latent is orders of magnitude cheaper to advance than the field, and the filter demonstrates the estimation half of that loop from the wall alone. Closed-loop actuation is outside the present evidence: the predictor is trained to forecast observed encounters within the training envelope, and predicting how an encounter would differ under intervention is a task its training does not address. The contribution is narrow by design, a controlled comparison and a frozen-filter characterisation on a single parametric flow, and what it establishes is bounded accordingly: a wall-estimable, forward-usable coefficient state, the ingredients that produce it, and the gust-intensity envelope over which a fixed sensing budget can follow it. The natural next steps are the noise model above and the same matched-estimator protocol on other parametric flows, to test how general the estimability ordering reported here actually is.

6. Conclusions

We asked which reduced-order state a sequential wall-pressure estimator can track through an extreme vortex-gust encounter, and reached a design rule with a division of labour and a working demonstration. On a pooled coefficient latent at matched dimension, observable supervision on the wake supplies both the instantaneous readability of the wake and an anisotropic latent geometry whose low-variance directions the forward rollout avoids, so a supervised state stays in distribution under prediction where an anti-collapse regulariser on a bare reconstruction, which is near-isotropic, does not; the multi-step predictive objective supplies the long-horizon, retrain-robust rollout stability that a single-step objective and an objective-free supervised encoder both lack. On this state we built a leakage-free ensemble Kalman filter that senses eight wall-pressure taps and never observes the vorticity field: it tracks the lift through the encounter, from an analysis closure of 0.74 to 0.84 for gust ratios of unity and above, where every static inverse, via any latent or straight from the full wall array, falls below the mean predictor by a ratio of three. Run identically on each family, the filter tracks the load through any nonlinear latent and fails only on the linear basis, so the extension of the envelope belongs to sequential estimation on a nonlinear state; what belongs to the wake-supervised predictive state is the boundary of that envelope, the calibration margin, and a tracked state that carries the wake at all. The recovery is a delay-coordinate reconstruction: the encounter is low-dimensional, so a few wall taps read over a short window place the state, one tap over a thirty-frame window matching eight taps at an instant in the static reconstruction, and the wake circulation the instantaneous pressure cannot see is recovered through the window, while the envelope closes where the strengthening gust raises the encounter’s effective dimension beyond what a fixed sensing budget can embed. The predictive coefficient state carries the most wall-recoverable physics of the three families we compared, though the reconstructive latent is the most recoverable in raw variance, a separation of energy from information at the wall. Three facts bound the result: the coefficient state’s recoverability advantage over a raw field latent is modest, the filter is under-dispersed at the strongest gusts and needs a calibrated noise model rather than inflation, and the mid-plane representation reaches a three-dimensional observability boundary at the highest gust strength. What the paper establishes is a wall-estimable, forward-usable reduced state and the ingredients that produce it, together with a filter that

demonstrates the state in the deployment the extreme-aerodynamic control literature has been pointing toward.

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Declaration of interests. The authors report no conflict of interest.

Data availability statement. The reduced-order-model and evaluation code, the configuration files, the v2.1 data-split manifest, and the analysis outputs that regenerate the tables and figures are deposited at <https://doi.org/10.xxxx/zenodo.PLACEHOLDER> under the MIT License (DOI to be finalised on acceptance). The processed per-encounter cache, or a representative subset sufficient to rerun the heavier steps, is available from the corresponding author. The raw direct numerical simulations were computed with the SOD2D solver (Gasparino *et al.* 2024) and are owned by the simulation collaborators.

Author contributions. [Authors to insert contributions, for example in CRediT format.]

Use of generative-AI tools. Generative-AI assistance (Anthropic Claude, accessed through the Claude Code command-line interface) was used during 2026 to help with data analysis, figure generation, and manuscript drafting. All numerical results were computed by the authors' own code and verified against the underlying data, and the authors take full responsibility for the content of the article.

Appendix A. Architecture, anti-collapse regularisation, and uncertainty quantification

Architecture detail

The encoder is a hybrid convolutional stem followed by a small vision transformer. Three downsampling convolutional stages map the single-channel 192×96 vorticity field to a 24×12 feature map at 256 channels, that is 288 image tokens, which a six-layer pre-norm transformer of width 256 with eight heads processes; the latent is read from a prepended class token through a one-layer projection whose output passes a BatchNorm, the layer the characteristic-function regulariser below requires. The encoder is unconditional and carries no information about the gust parameters. The training predictor advances the pooled 32-vector directly: its function class is a six-layer autoregressive transformer of width 384 with sixteen heads, rotary temporal positions and a causal mask, mapping a short history of 32-vectors to the next 32-vector, with no spatial map or tiling anywhere in the loop. Training minimises an open-loop rollout loss over eight steps, with no teacher forcing, against online encoder targets that are detached (no exponential-moving-average target network), plus the regulariser below, with AdamW and bf16 mixed precision. The predictor advances the latent from its own history alone, with no gust input, so the gust parameters $\mathbf{c} = (G, D, Y)$ enter nowhere in the model. Two downstream forecast operators are fitted afterwards on each frozen latent, identically across families. The forecast merit of table 3 uses a common residual U-Net operator, chosen for numerical stability across every latent geometry: a matched transformer of the training predictor's class, refitted on the frozen latents, forecasts the predictive latent well but is unstable on latents outside its design, collapsing to a negative merit on the beta-VAE control, and so cannot serve as the common yardstick. The ensemble filter of §3.3 uses a matched autoregressive transformer on the flagship's own pooled trajectories, where it is stable; and the flagship's own co-trained predictor, the as-built reduced-order model, is reported alongside the common-operator merit. The instability is specific to latents outside the transformer's design: refitted on the pooled predictive, supervised and reconstructive-autoencoder latents the transformer forecasts them comparably to the residual U-Net, and on the predictive flagship it in fact leads the pooled families at 0.655, above the objective-free control; but on the reference latents (the beta-VAE and the two Fukami autoencoders) it diverges to large negative merit. This operator dependence is the reason the common-operator merit is read as a level rather than a fine ranking between the two supervised states, and the reason the residual U-Net, stable on every family, is the yardstick of record.

Anti-collapse regularisation

Because the encoder and predictor are trained end to end on latent-space prediction with no decoder, the latent is held away from a collapsed solution by an explicit regulariser rather than by a reconstruction term. We use SIGReg, the characteristic-function (Epps–Pulley) regulariser of the from-pixels joint-embedding line (Maes *et al.* 2026; Balestrieri & LeCun 2025), with M random one-dimensional projections of the latent and a fixed weight $\lambda_S = 0.02$ in the loss, the value used in every run reported here. The same regulariser has since been adopted independently for biosignal time series, where it is likewise reported to drive the embedding toward isotropy so that a structuring objective is needed alongside it (Broustail *et al.* 2026), consistent with the supervision-anisotropy mechanism of §4.2. As a safeguard the participation ratio of the latent is monitored

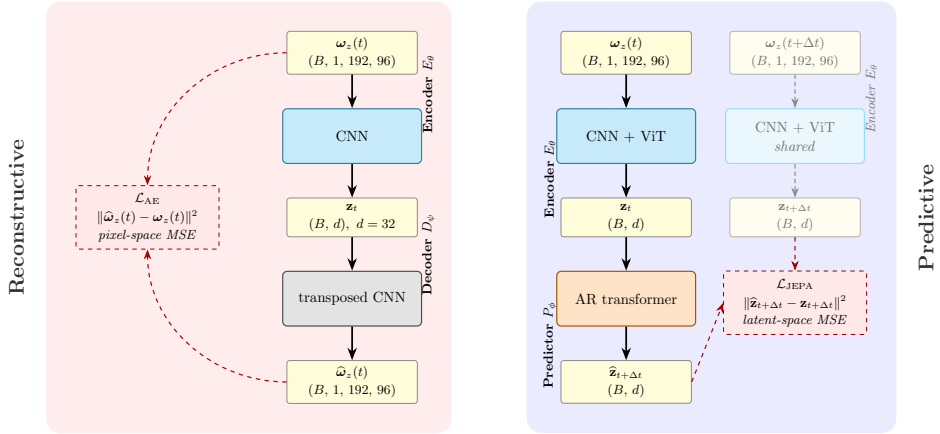


Figure 11. Predictive versus reconstructive training. Left: the reconstructive autoencoder encodes the field at t , decodes it, and incurs a field-space error. Right: the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss and no decoder during training. This is the contrast the matched-predictor protocol of §3.2 evaluates; the visualisation decoder is trained only afterwards, on the frozen encoder.

every thousand iterations, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) if it indicates collapse while the parameter probe is still uninformative; on the runs reported here the participation ratio stays healthy and the fall-back does not trigger.

Uncertainty quantification

Every held-out coefficient of determination and mean absolute error in the paper carries three independent uncertainty signals, following the reporting protocol fixed with the split. The first is a bootstrap confidence interval from 2000 resamples over the held-out test encounters, which is the dominant source of scatter at the present sample size ($n = 42$ for test_b, $n = 40$ for test_c) and is wide for the individual wake observables, as noted in §4.1. The second is the variance across three encoder retrains from independent random seeds, reported as the standard deviation in the objective-times-architecture controls of §4.2. The third is a five-fold cross-validation over cases on the read-out (probe) step, which guards against a probe that overfits the particular held-out split. Together these approximate what a full five-fold case-level cross-validation of the entire pipeline would buy, at a fraction of the training cost; a single split with bootstrap intervals is the community standard for this class of study.

Training-set fit

The predictive latent attains the highest in-distribution training fit of the forecast rollout among the families, but this is an in-distribution number, not the held-out result: the family ordering is established by the held-out representational closure of table 3, and the held-out forecast mean is markedly lower than the training fit, as it must be. We therefore do not tabulate the training-set fit, which would invite reading an in-distribution number as a result.

Paired tests on the estimation endpoints

Table 8 reports the pre-registered per-encounter paired family behind the tracking and envelope claims of §4.4 and §4.5. Pairing cancels the encounter-to-encounter difficulty that widens marginal intervals, and clustering by case respects the shared shedding history of a case’s encounters.

Topology of the encoded encounter

A coordinate-free check on the geometric mechanism of §4.2: Vietoris–Rips persistence of the encoded trajectories, per family, with a generator declared significant when its persistence exceeds a noise floor set at a fraction of the cloud diameter, after per-family standardisation so that coordinate scale cannot carry the result. On the gusted held-out encounters the supervised states keep a single clean H_1 cycle in about half the

Table 8. Paired per-encounter tests for the estimation endpoints, on the frozen filter records. The family was pre-registered before any statistic was computed: the primary family F1 compares the filter analysis against the static single-frame recovery and F2 against the field-free open-loop forecast, on the endpoints $\{C_L, E_w\}$ in the impact and relaxation windows, one-sided with the filter pre-registered as better; ΔR^2 is the median per-case mean improvement, the p is a case-level Wilcoxon signed-rank, and Holm corrects within each family. On the pooled test_b encounters the filter beats the open-loop forecast decisively on the load but not the static recovery, which is accurate at the moderate gust strengths test_b samples; the filter's advantage over the static inverse concentrates at strong gusts, where the test_c annex (characterisation only, $n = 8$ cases) is decisive on the load in both windows. The wake enstrophy through the filter does not beat the static recovery at the boundary, an honest null.

Family	Endpoint	Window	median ΔR^2	Wilcoxon p	Holm p
F1 (vs static)	C_L	impact	0.16	0.161	0.483
F1 (vs static)	C_L	relax.	0.53	0.116	0.465
F1 (vs static)	E_w	impact	-25.15	0.986	1.000
F1 (vs static)	E_w	relax.	-2.47	0.839	1.000
F2 (vs open loop)	C_L	impact	9.43	0.001	0.003
F2 (vs open loop)	C_L	relax.	44.84	0.001	0.003
F2 (vs open loop)	E_w	impact	17.39	0.246	0.246
test_c annex	C_L	impact	1.11	0.0078	–
test_c annex	C_L	relax.	6.41	0.0039	–

Table 9. Preprocessing robustness of the representational wake closure, established on the previous-generation spatial encoders ($d = 64$, the 85-case v2.1 split) and retained as evidence that the amplitude clip does not shape the endpoint; the pipeline is unchanged in the present dataset revision, and the probe protocol is identical. Held-out test_b wake-enstrophy R^2 from a frozen-latent ridge probe (no retraining), recomputed under three training-only amplitude-clip treatments of the raw ω_z field. The predictive latent leads the wake under every treatment; the advantage keeps its sign and approximate magnitude.

Family	no clip	per-encounter	training-global
predictive (JEPA)	0.75	0.80	0.84
reconstructive (Fukami)	0.12	0.10	0.08
linear (POD)	-0.27	-0.05	0.00

encounters (0.64 for the predictive and 0.52 for the objective-free supervised state) where the anti-collapse reconstruction fragments (0.10); on the single-period no-gust control the same separation holds (0.62 against 0.00), and it survives whitening, so the fragmentation is a property of the reconstructive encoding rather than of any coordinate scale. The linear basis keeps the cleanest no-gust loop of all (0.62), consistent with a linear projection of a smooth field; the claim is that the anti-collapse reconstruction fragments, not that only the predictive latent carries a loop.

Preprocessing robustness

The wake enstrophy squares the vorticity, so the per-encounter high-percentile amplitude clip of the ω_z preprocessing could in principle shape the primary endpoint. We test this on the frozen encoders, with no retraining, by recomputing the readout-frame wake-enstrophy target under three training-only clip treatments of the raw field and re-fitting the frozen-latent-to-wake ridge probe per family: no amplitude clip, the production per-encounter $p_{99,99}$ clip, and a single leak-free $p_{99,99}$ threshold pooled over the training encounters. The predictive latent leads the wake on test_b under all three (table 9), with the largest shift in the predictive-minus-best-baseline advantage relative to the production clip being 0.07 in R^2 . The leak-free training-global clip gives the largest predictive advantage, so the per-encounter clip is not the source of the wake result.

Appendix B. Sparse wall-pressure state estimation

At deployment the vorticity field is not available; only the wall pressure is. This appendix records the sensing conventions behind §4.3 and §4.4 and the placement-robustness checks that the main text summarises. The input throughout is the span-averaged pressure on the 192 surface taps; the recovery estimator is selected per family by encounter-grouped cross-validation among a kernel-ridge regressor on the flattened window, a multilayer perceptron, and a recurrent network over the window sequence, and the recovered state is read by the same frozen probes as everywhere else.

Placement, and why it does not drive the ordering

Two placements are used, deliberately. The filter and the recovery of table 5 sense at each family’s own model-conditioned taps, the greedy selection that best exposes that family’s latent, because the estimator comparison should give every family its best sensing. The delay-trade grid of §4.3.1 instead uses a single target-blind array, the QR-pivot optimum of the wall-pressure modes (Manohar *et al.* 2018), nested so that every tap count is a prefix of the next, because a sensor-budget claim must not depend on a placement tuned to any model. The two meet where their sensing coincides: the grid’s eight-tap, thirty-frame cell rerun on the model-conditioned taps reproduces the table’s recovery exactly (0.656). The energy-versus-information split of §4.3 is also placement-robust: under the shared target-blind array with the kernel-ridge estimator alone, the reconstructive state remains the most recoverable in variance (0.775 against 0.667 for the predictive and 0.500 for the linear basis) and the predictive state retains the most recoverable physics (0.523 against 0.363 and 0.391), so neither ordering is an artefact of the model-conditioned placement.

What the window recovers that the instant cannot

At a single frame the wall is blind to the wake-circulation direction of the predictive state (§4.3); over the thirty-frame window the blind coordinate becomes partially recoverable (0.33 held-out, from near zero at one frame), and the wake enstrophy read through the recovered state reaches 0.40. The window, not the tap count, is what carries this: the same quantities barely move as taps are added at a fixed single frame. This is the delay-coordinate content of the sensing result, quantified in the grid of §4.3.1; it is a statement about the static reconstruction, and the filter inherits the sequential version of it through its assimilation window rather than through any explicit delay embedding.

The recovered field

Decoding the pressure-recovered state through each family’s diagnostic decoder makes the recovery visible in physical space (figure 12). The comparison within each row, recovered field against that family’s own decode ceiling, isolates the pressure step from the decoder’s reconstruction limit: the loss from ceiling to recovery is the part attributable to the eight-tap window, and the panels show it concentrating in the wake structures rather than in the near-body field, consistent with the wall reading the force-carrying directions best (§4.3).

The out-of-distribution boundary from the sensing side

At the $|G| = 4$ boundary every static approach fails on the load, the via-latent recovery at -0.33 and the direct no-latent regression at 0.08 from the same taps (0.14 with the full array), while the filters still track it (table 6). The boundary is the same three-dimensional observability limit the field reaches (§2.1): the mid-plane state no longer determines the dynamics there, so a static inverse has nothing to invert, and only the accumulated pressure history carries the load through.

Delay-coordinate knobs

The window sweeps of §4.3.1 keep the delay stride at the cache cadence ($\Delta t = 0.05 c/u_\infty$), and this choice is a convention rather than a tuning: the first minimum of the time-lagged mutual information of the tap signals (Fraser & Swinney 1986) sits near 29 frames, about one shedding period, so the thirty-frame window spans roughly one decorrelation time of the wall pressure at the cadence used. The window length itself is selected by the measured recovery trade rather than by a false-nearest-neighbours criterion (Kennel *et al.* 1992), which we regard as a cross-check and not a design input: the delay-embedding guarantees are generic, wall pressure is a physical and non-generic observable strongly coupled to the near-body vorticity, and the operative evidence is the measured recovery, with the theorem as the organising principle rather than a proof that a given tap count suffices.

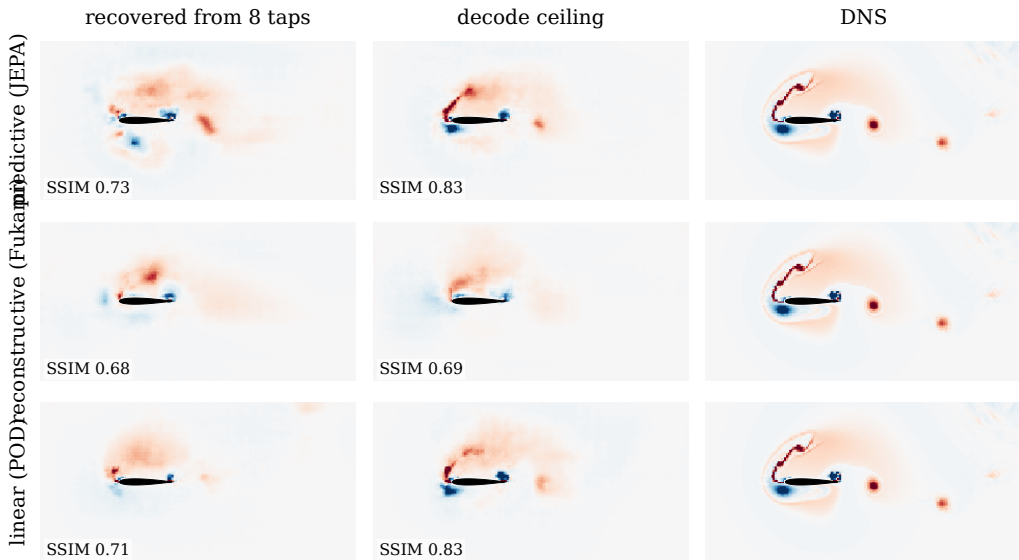


Figure 12. Flow recovered from eight wall-pressure taps on the representative held-out encounter of figure 8, at the impact-side frame, one row per family. Columns: the field decoded from the pressure-recovered coefficient state, the decode ceiling (the same decoder applied to the encoded state, no pressure step), and the simulation; per-panel structural similarity annotated. The comparison is within each row, since the decoders' ceilings differ (§5.3); the ceiling-to-recovered gap is the cost of the pressure step.

Appendix C. Supplementary figures

The figures below are referenced from the Results and Discussion and collected here to keep the main-text figure sequence focused on the narrative steps: the decoded vorticity fields behind the decode-similarity column of table 3 (figure 13) and the decodability-versus-estimability trade behind §5.3 (figure 14).

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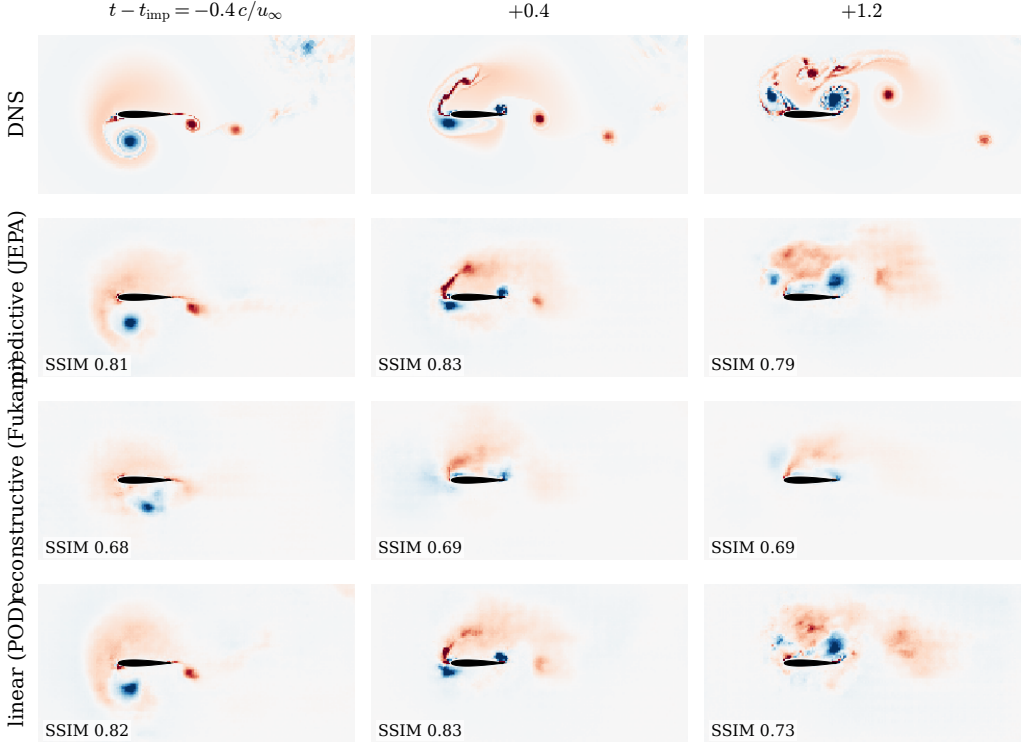


Figure 13. Decoded vorticity through the encounter on the representative held-out test_b case ($|G| = 1.5$, the same encounter as figure 8). Rows: the simulation and the fields decoded from the encoded coefficient state of the predictive, reconstructive, and linear families; columns: a pre-impact, an impact, and a relaxation frame. Each decode is produced by the same diagnostic decoder that defines the decode-floor similarity of table 3, trained on the frozen encoder; the per-panel structural similarity is annotated. The pooled decode is soft by construction (§5.3); the comparison is between rows at matched frames, and no family’s decode is part of any training loss.

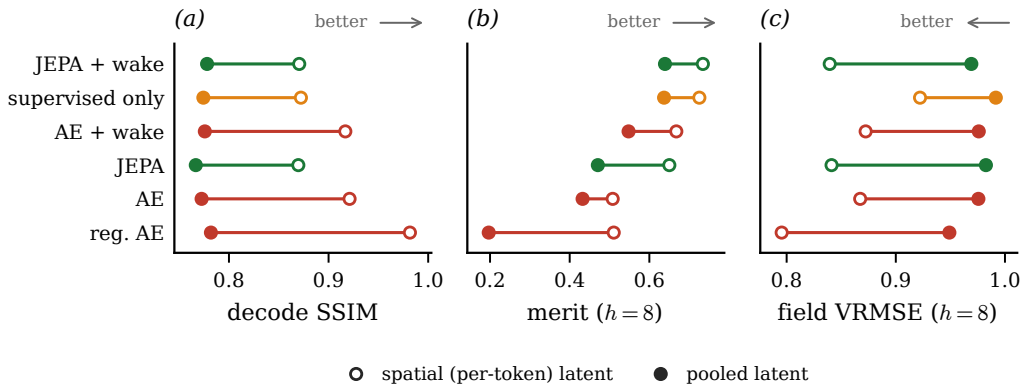


Figure 14. The decodability-versus-estimability trade of §5.3, quantified. Each panel connects the spatially resolved latent (open marker) to its pooled coefficient state (filled marker), per family, for the decode similarity, the matched-predictor merit, and the field forecast error. Pooling costs field fidelity in every family; the coefficient state is the price of a state a wall-pressure filter can track.

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